

BFW STEAM HEATER VER 1

Operator Training Simulator · E-101

Chulalongkorn University · Department of Chemical Engineering

HAZOP REPORT

Hazard and Operability Study of the E-101 BFW steam heater

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Teaching document generated from the HAZOP tab of the E-101 simulator. It is not a study of a real plant.

Rev.	Date	Revision status	Prepared	Checked	Approved	Remark
A	03-Oct-26	Issued for teaching	A. Soottitantawat			

HAZOP CLOSE-OUT SUMMARY

Project: BFW Steam Heater Ver 1 · E-101 Operator Training Simulator · Document BFWH-V1-SA-HZP-001 rev A · 28 actions, all open at issue. The action party fills "Action taken" and the HAZOP leader changes the status to CLOSED when the close-out sheet (Appendix 3) is signed.

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N1 BFW supply: TK-101 → P-101A/B → recycle FV-101B	R01	N1.1, N1.11, N6.11	Auto-start P-101B on a P-101A trip or low discharge pressure (PI-106 transmitter, recommended addition), completing inside the FSLL-101 delay; test the spare start monthly.	Instrument / Operations		OPEN	Risk 12 High
N1 BFW supply: TK-101 → P-101A/B → recycle FV-101B	R02	N1.1, N1.2, N1.3, N1.4, N1.6	Add level transmitter LT-101 (recommended addition) on TK-101 with a low-level alarm and a low-low pump trip, and a high-level alarm for overfill.	Instrument		OPEN	Risk 12 High
N1 BFW supply: TK-101 → P-101A/B → recycle FV-101B	R03	N1.5	Provide a fixed minimum-flow line with restriction orifice from the pump discharge to TK-101, independent of FV-101B, so a blocked outlet downstream of E-101 cannot deadhead the running pump.	Process		OPEN	Risk 6 Medium
N1/N2 BFW supply: TK-101 → P-101A/B → recycle FV-101B / BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	R04	N1.7, N2.9	Confirm a check valve on each pump discharge and on the BFW line to the boiler house; add them to the inspection plan.	Mechanical		OPEN	Risk 4 Low
N2 BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	R05	N2.3, N2.4	Fit a thermal relief valve on the E-101 tube side (recommended addition), or lock the procedure so the steam is isolated and the shell cooled before the BFW is blocked in.	Process		OPEN	Risk 8 Medium
N2 BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	R06	N2.2, N2.5, N2.7	Add a deviation alarm between TT-101 (control) and TIAH-108 (independent) so a transmitter reading low or frozen is seen before the boiler house is affected.	Instrument		OPEN	Risk 9 Medium
N1/N2 BFW supply: TK-101 → P-101A/B → recycle FV-101B / BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	R07	N1.1, N1.2, N2.1, N2.2, N2.5	Agree the interface with the boiler house: shared alarms for FIC-101 LO and TIAH-108 HI, a call-out procedure, and include loss of BFW and hot BFW from E-101 in the boiler-house HAZOP.	Operations / Process		OPEN	Risk 12 High
N3/N4 Steam supply: header → XV-100 → FT-104 → FV-104 / E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	R08	N3.1, N3.3, N4.5, N4.8	Confirm the E-101 shell design pressure is at least the highest credible header pressure (144 °C saturation at 300 kPa g in the fault library) and that PSV-100 is set at or below it and sized for FV-104 fully open at that header pressure and for the external-fire case.	Process		OPEN	Risk 10 High
N3 Steam supply: header → XV-100 → FT-104 → FV-104	R09	N3.2, N3.4	Header-pressure response procedure with the upstream flash-steam plant for PI-105 LO/HI; consider a pressure-control station on the header so E-101 never sees more than its design pressure.	Operations / Process		OPEN	Risk 9 Medium
N3 Steam supply: header → XV-100 → FT-104 → FV-104	R10	N3.1, N3.6, N3.7	Add FV-104 position feedback ZT-104 (recommended addition) with a deviation alarm between FIC-104 output and valve position (stuck or failed valve).	Instrument		OPEN	Risk 9 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N4 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	R11	N4.2	Apply the tube-rupture (two-thirds) rule: BFW side up to 1030 kPa g at pump shut-off. If the shell design pressure is below two-thirds of the tube-side design pressure, size PSV-100 for the tube-rupture flow of water and flashing water.	Process		OPEN	Risk 10 High
N4/N5 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100 / Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	R12	N4.2, N4.3, N5.1, N5.3	Add a shell hotwell level transmitter LT-100 (recommended addition) with a high alarm below the tube bottom (12 % of the shell) and a high-high level trip closing XV-100 before steam can meet a flooded shell.	Instrument / Process		OPEN	Risk 12 High
N4 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	R13	N2.6, N4.1, N6.3	Put a continuous vent in service (rev B RO-101) or the automatic air vent with vent-line temperature (rev C AAV-101, TI-102, TAL-102): gas blanketing gives no alarm for a long time.	Process		OPEN	Risk 8 Medium
N4 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	R14	N3.5, N4.4	Confirm the shell is rated for full vacuum or that VB-101 is sized for the fastest condensation after a trip; add VB-101 to the preventive-maintenance and test plan.	Mechanical		OPEN	Risk 9 Medium
N5 Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	R15	N2.6, N5.1, N5.3, N5.4, N5.5, N5.7	Trap management: regular trap survey (ultrasonic or temperature), Y-101 cleaning after any maintenance, and confirm ST-101 capacity at low shell differential pressure (start-up and near-vacuum operation).	Mechanical / Operations		OPEN	Risk 12 High
N5 Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	R16	N5.2, N5.6	Replace local gauge PG-110 by a return-header pressure transmitter with a high alarm (recommended addition); route shell drain V-105 to a flash pot or blowdown pit, not to an open sewer.	Process / Instrument		OPEN	Risk 9 Medium
N4/N5 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100 / Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	R17	N4.7, N4.8, N5.6	Route V-104, PSV-100, V-103 and V-105 discharges to a safe location; fit a silencer on the vent; warning signs and hearing-protection zone around the vent and trap set.	Mechanical / Safety		OPEN	Risk 9 Medium
N6 Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	R18	N6.2, N6.3, N6.7	Start-up procedure hold points: BFW in AUTO first, vent V-104 open, warm-up with FIC-104 below 10 % (the model gives water hammer at 10 % in a cold shell, none at 5 %) until the shell is warm, vent until dry steam blows, then cascade.	Operations		OPEN	Risk 12 High
N6 Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	R19	N2.4, N6.5, N6.6	Shutdown procedure: steam off and shell depressured before the BFW is stopped or blocked in; open V-105 only after the shell pressure has fallen; keep V-104 open during lay-up.	Operations		OPEN	Risk 8 Medium
N6 Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	R20	N6.1, N6.9	Trip-bypass management for FSLL-101, TSHH-108 and PSHH-100: permit, time limit, MOC, bypass status shown on the DCS; proof-test the XV-100 trips at the interval set by LOPA.	Operations / Instrument		OPEN	Risk 8 Medium
N6 Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	R21	N3.5, N6.8	Restart after any trip: re-vent the shell (air drawn in through VB-101) before the cascade is put back; consider "V-104 open" as a start permissive.	Operations / Instrument		OPEN	Risk 9 Medium

Section / Node	Action No.	Raised from	Recommendation	Action by	Action taken	Status	Remark
N6 Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	R22	N6.10, N6.11	Confirm the fail position of FV-101B (not shown on the P&ID) and that XV-100 closes on loss of air or power; document the fail positions in the cause-and-effect chart.	Instrument		OPEN	Risk 9 Medium
all all	R23	N3.3, N4.2	LOPA for the High rows that rely on PSHH-100, FSLL-101 and PSV-100 (shell overpressure, tube rupture, loss of BFW, water hammer).	Process safety		OPEN	Risk 10 High
N3 Steam supply: header → XV-100 → FT-104 → FV-104	R24	N3.8	Show a calculated superheat TI-100 – T _{sat} (PI-100) on the DCS; tell the upstream plant when it exceeds a few degrees.	Instrument		OPEN	Risk 4 Low
N1/N2 BFW supply: TK-101 → P-101A/B → recycle FV-101B / BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	R25	N1.10, N2.8	Water-treatment control: BFW hardness and oxygen-scavenger sampling programme; closed sampling point and PPE for treatment chemicals.	Operations		OPEN	Risk 6 Medium
N3 Steam supply: header → XV-100 → FT-104 → FV-104	R26	N3.9	Confirm a drip leg and trap on the steam branch upstream of XV-100, and warm the line with its drains open before admitting steam.	Mechanical / Operations		OPEN	Risk 8 Medium
N4 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	R27	N4.6	Condensate quality: oil-in-condensate check or monitor on the return to the boiler house; find and stop the oil source in the flash steam.	Operations		OPEN	Risk 6 Medium
N4/N5 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100 / Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	R28	N4.9, N5.9	Maintenance isolation procedure for E-101: positive isolation (blinds) of steam, BFW and condensate, drain and cool before opening, confined-space permit for shell entry, lift plan for bundle pulls.	Operations / Maintenance		OPEN	Risk 9 Medium

HAZOP REPORT

Hazard and Operability Study of the E-101 flash-steam BFW heater
BFW Steam Heater Ver 1 · E-101 Operator Training Simulator

	Role	Name	Signature	Date
Prepared by	HAZOP scribe			
Reviewed by	HAZOP leader (chair)			
Approved by	Plant / project manager			

This report records a teaching HAZOP of the heater drawn in the simulator. Names, signatures and dates are filled in by the class team that runs the study; the risk matrix is an example for teaching, not a corporate matrix.

CONTENTS

HAZOP CLOSE-OUT SUMMARY	2
Executive summary	7
1 Introduction	8
1.1 Purpose	8
1.2 The unit	8
1.3 Objectives	8
2 Scope	8
2.1 Nodes	8
2.2 Node map	9
3 Methodology	10
3.1 References	10
3.2 The four phases	10
3.3 Guide words and parameters	10
3.4 Risk ranking	11
4 HAZOP sessions	12
4.1 Schedule	12
4.2 Team	12
4.3 Documents reviewed	12
4.4 Nodes reviewed and guide-word coverage	12
5 Findings and recommendations	13
5.1 Risk profile	13
5.2 Recommendations	13
5.3 Dynamic check of selected deviations on the simulator	14
5.4 Follow-up and close-out	15
Appendix 1 · HAZOP worksheets	16
Node N1 · BFW supply: TK-101 → P-101A/B → recycle FV-101B	16
Node N2 · BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	18
Node N3 · Steam supply: header → XV-100 → FT-104 → FV-104	20
Node N4 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	22
Node N5 · Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	23
Node N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	24
Appendix 2 · HAZOP documents (drawings with node boundaries)	25
A2-1 · HAZOP node map	25
A2-2 · P&ID-001 rev C (screen drawing used for the study)	26
A2-3 · BFWH-V1-PID-001 rev D (alternative EPC drawing)	27
Appendix 3 · HAZOP action close-out sheets	28

Executive summary

A HAZOP of the E-101 flash-steam boiler-feed-water heater, from the BFW tank TK-101 and pumps P-101A/B to the delivery to the boiler house, including the flash-steam branch with XV-100 and FV-104, the shell with its vent, vacuum breaker and relief, the condensate trap set, the SIS and the start-up and shutdown procedures, was carried out with the IEC 61882 guide-word method. The heater was divided into **6 nodes**; **58 deviations** were recorded and **28 recommendations** raised.

Nodes	Deviations	Recommendations	Very high	High	Medium	Low
6	58	28	0	5	47	6

Risk is ranked with the existing safeguards credited (Section 3.4). High-ranked deviations are carried to the LOPA (LOPA and SRS of SIF tabs).

Key findings (dynamic check on the simulator)

- **N1 loss of BFW:** FSLL-101 closes XV-100 at 00:07. Before the trip the shell climbs to 193 kPa g (no water left to condense the steam); after it the shell falls to –3 kPa g as the cold tubes condense what is left, so VB-101 admits air and the restart must vent (R14, R21). FSLL-101 protects the heater, not the boiler house: the loss of BFW itself needs the spare auto-start and the boiler-house interface (R01, R07).
- **N2 / N3 more heat:** with FV-104 wide open the true outlet only reaches 74.5 °C and with the BFW cut to 60 m³/h about 82.9 °C, below TSHH-108 (95 °C). The header pressure, not the valve, limits the temperature; TSHH-108 is the backstop for combinations the model does not reach (for example very low flow above the FSLL-101 setting with a high header).
- **N2 less BFW:** when the flow is cut to 60 m³/h the cascade throttles the steam so hard that the shell dips to –3 kPa g; the trap cannot discharge below the return-header pressure and the hotwell rises to 17 % (tubes from 12 %). A load change can flood the shell without any trap fault (R12, R15).
- **N3 shell overpressure needs two failures:** PSHH-100 trips at 00:55 but the shell still peaks at 255 kPa g because of the 3 s delay. PSV-100 must be set at or below the shell design pressure and sized for this case (R08, R23).
- **N4 gas blanketing:** no alarm at all in 30 min; PI-100 creeps from 105 to 172 kPa g and FV-104 opens to 100 % while TT-101 stays near set point. The only early signs are a valve and a pressure that drift, so the continuous vent (R13) matters more than an alarm.
- **N4 tube leak:** no alarm; the steam rises to 10,841 kg/h and the hotwell to 14 %, just into the tubes. Without a level instrument (R12) a leak is found only by a condensate balance.
- **N5 trap blocking:** first alarm TIC-101 LO at 08:36, long after the hotwell has started to flood (67 % after 30 min, outlet down to 40.4 °C). Left longer it ends in water hammer (the precomputed 1 h run bangs at 34:43). A shell level alarm (R12) would warn before either.

Actions by action party

Action party	Number	Action No.
Operations	12	R01, R07, R09, R15, R18, R19, R20, R21, R25, R26, R27, R28
Instrument	10	R01, R02, R06, R10, R12, R16, R20, R21, R22, R24
Process	9	R03, R05, R07, R08, R09, R11, R12, R13, R16
Mechanical	5	R04, R14, R15, R17, R26
Safety	1	R17
Process safety	1	R23
Maintenance	1	R28

An action shared by two parties (e.g. Instrument / Operations) is counted under both.

1 Introduction

1.1 Purpose

The HAZOP identifies how E-101 can deviate from its design intent, what the deviations lead to, whether the existing safeguards are adequate, and what should be changed. It also finds operability problems: deviations that do not harm people but upset the boiler house or waste flash steam. The study feeds the safety layers, LOPA and SRS of the SIFs, and the cause and effect diagram BFWH-V1-IN-CE-001.

1.2 The unit

E-101 is a horizontal shell-and-tube heater. Boiler feed water from TK-101 is pumped by P-101A (P-101B spare) under FIC-101 (FV-101A, with the minimum-flow recycle FV-101B) through 100 tubes, one pass, and is heated from 18 to 70 °C (6.04 MW). Low-pressure flash steam from a 200 kPa g header condenses in the shell at about 105 kPa g; TIC-101 sets the steam flow FIC-104 (FV-104) in cascade. Non-condensable gas is vented at V-104 at the far end of the shell; VB-101 breaks vacuum and PSV-100 relieves overpressure. Condensate leaves through strainer Y-101 and trap ST-101 to a 30 kPa g return header. SIS I-100 closes XV-100 on low BFW flow (FSLL-101), high outlet temperature (TSHH-108), high shell pressure (PSHH-100) or the ESD push-button.

1.3 Objectives

- Identify hazards to people, the environment and the plant from deviations of each node.
- Identify operability problems and their effect on the boiler house and the flash-steam plant.
- Judge the existing safeguards and rank the risk with the example matrix.
- Raise recommendations and pass the high-ranked scenarios to LOPA.
- Check selected deviations dynamically on the simulator model (Section 5.3).

2 Scope

Boundaries: scope E-101 with TK-101, P-101A/B, FIC-101 and the recycle, the steam branch from the flash-steam header (PI-105) through XV-100 and FV-104, the shell with its vent, vacuum breaker and relief, and the condensate trap set to the return header; the SIS (XV-100 trips). Excluded: the upstream flash-steam plant, the condensate return system beyond the header and the boiler house (their interfaces are recorded as causes and consequences). Objectives, team, risk matrix.

2.1 Nodes

Node	Description	Design intent	Design conditions	Equipment / instruments
N1	BFW supply: TK-101 → P-101A/B → recycle FV-101B	Transfer treated BFW at 18 °C from TK-101 with P-101A (P-101B spare) at about 662 kPa g discharge; 10.7 m³/h returns to TK-101 through FV-101B so the running pump always has a minimum flow.	18 °C; discharge 662 kPa g; shut-off 1030 kPa g; 110.7 m³/h pumped	TK-101, P-101A/B, suction and discharge piping, FV-101B recycle <i>PI-106, TI-103, FIC-101 (flow seen by the pumps)</i>
N2	BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	Deliver 100 m³/h of BFW through the 100 tubes of E-101 (FV-101A, fail open), heat it from 18 to 70 °C and send it to the boiler house deaerator at about 602 kPa g.	100 m³/h; 18 → 70 °C; tube ΔP 60 kPa; outlet 602 kPa g	FV-101A, FT-101, E-101 tubes (one pass), outlet line to the deaerator <i>FIC-101, FSLL-101, TI-103, PDI-109, TT-101 / TIC-101, TIAH-108 / TSHH-108</i>
N3	Steam supply: header → XV-100 → FT-104 → FV-104	Take 9,935 kg/h of flash steam from the header at about 195 kPa g and throttle it with FV-104 (fail closed) under FIC-104, whose set point comes from TIC-101; XV-100 isolates the steam on any trip.	Header 200 kPa g (133.9 °C sat.); 9,935 kg/h; FV-104 50 % open	Steam branch, XV-100 (SIS), FT-104, FV-104 <i>PI-105, FIC-104, TIC-101 (cascade master), I-100 (SIS)</i>
N4	E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	Condense 9,935 kg/h of steam at about 105 kPa g (121.3 °C) on the whole tube surface, keep non-condensable gas vented, and keep the shell between atmospheric pressure and its design pressure.	105 kPa g; 121.3 °C; duty 6.04 MW; hotwell 5 % (tubes from 12 %)	E-101 shell, V-104 air vent (rev B RO-101, rev C AAV-101), VB-101, PSV-100 <i>PI-100, PSHH-100, TI-100 (TI-102 / TAL-102 with rev C)</i>

Node	Description	Design intent	Design conditions	Equipment / instruments
N5	Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	Remove 9,935 kg/h of condensate at about 119 °C to the return header at 30 kPa g, keep the hotwell at 5 % of the shell (below the tubes at 12 %) and hold steam in the shell.	119 °C; 30 kPa g return (107.4 °C saturation); hotwell 5 %	V-101, Y-101 strainer, ST-101 trap, V-102, V-103 bypass, V-105 drain to sewer <i>None on the P&ID (local PG-110 return pressure, TG-107 condensate temperature)</i>
N6	Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	Bring E-101 from cold to 70 °C without water hammer or air blanketing, shut it down and restart it safely, and survive loss of air or power.	Start permissives: SIS reset, BFW ≥ 50 m ³ /h, FIC-104 in MAN ≤ 10 %, trap line open	Whole unit; SIS start permissives and trips <i>SIS screen (XV-100), FIC-104 in MAN, V-104, V-105, V-101 / V-102</i>

2.2 Node map

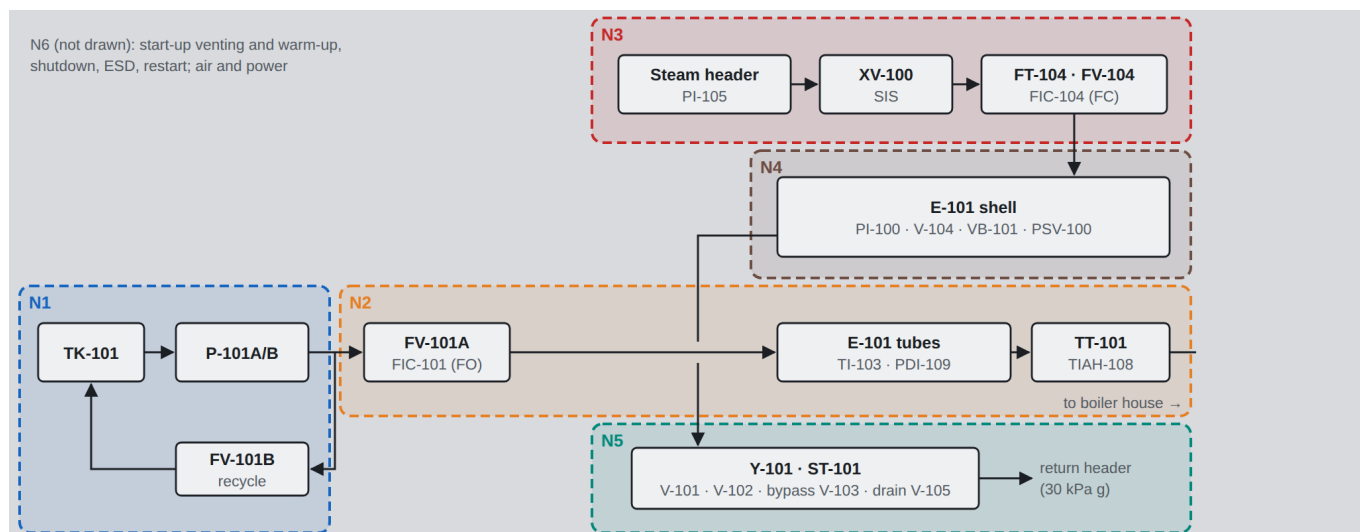


Figure 1. E-101 divided into HAZOP nodes (N6, the procedures node, covers the whole heater).

3 Methodology

HAZOP is a structured, team-based examination of a defined system. Each part of the design is compared with its **design intent**; **guide words** combined with process **parameters** generate **deviations**; for each meaningful deviation the team records causes, consequences, existing safeguards and, where the risk is not acceptable, recommendations. It finds hazards (harm to people, environment, assets) and operability problems (the heater cannot deliver hot BFW, or cannot be run as intended).

3.1 References

Reference	What it contributes
IEC 61882:2016 <i>Hazard and operability studies (HAZOP studies) – Application guide</i>	Definitions, the four phases (definition, preparation, examination, documentation and follow-up), guide words, team roles, recording
CCPS <i>Guidelines for Hazard Evaluation Procedures</i>	Worksheet format, node selection, consequences judged without safeguards, risk ranking, link to LOPA
IEC 61511 (functional safety, SIS)	How HAZOP scenarios feed LOPA and SIL targets for the XV-100 trips FSLL-101, TSHH-108 and PSHH-100
ISO 31000 / owner's risk matrix	Risk criteria used to decide whether safeguards are adequate
TEMA / ASME Section VIII, API 520/521	Shell-and-tube exchanger design pressures, the tube-rupture ("two-thirds") rule, relief sizing for PSV-100 and vacuum design of the shell
Thailand: Department of Industrial Works regulation on hazard identification, risk assessment and risk-management plans (B.E. 2543)	Legal requirement to identify hazards (HAZOP is an accepted method) and to prepare risk-management plans for factories
OSHA 29 CFR 1910.119 (PSM), where applicable	Process hazard analysis revalidated at least every five years (a steam heater is usually covered only as part of a larger process)

3.2 The four phases

- **Definition:** scope E-101 with TK-101, P-101A/B, FIC-101 and the recycle, the steam branch from the flash-steam header (PI-105) through XV-100 and FV-104, the shell with its vent, vacuum breaker and relief, and the condensate trap set to the return header; the SIS (XV-100 trips). Excluded: the upstream flash-steam plant, the condensate return system beyond the header and the boiler house (their interfaces are recorded as causes and consequences). Objectives, team, risk matrix.
- **Preparation:** collect the P&ID, PFD, heat and material balance, trip (cause-and-effect) list, alarm list, procedures and datasheets; divide the unit into nodes; write each node's design intent; choose parameters and guide words.
- **Examination:** node → design intent → parameter → guide word → is the deviation meaningful? → causes → consequences assuming the safeguards fail → existing safeguards → rank the risk → recommend → next guide word. Every meaningful deviation is recorded, including those with adequate safeguards (**full recording**), so the reasoning can be audited.
- **Documentation and follow-up:** worksheets, recommendation register with owners and dates, close-out, revalidation after changes (MOC) or periodically.

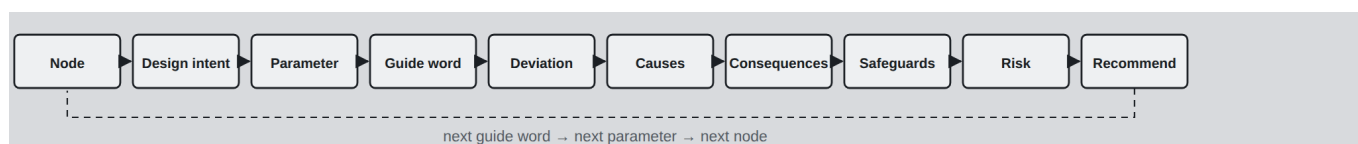


Figure 2. HAZOP examination loop.

3.3 Guide words and parameters

Guide word	Meaning (IEC 61882)
NO / NOT	Complete negation of the intent
MORE	Quantitative increase
LESS	Quantitative decrease
AS WELL AS	Qualitative increase: something extra present
PART OF	Qualitative decrease: only part of the intent achieved
REVERSE	Logical opposite of the intent
OTHER THAN	Complete substitution: something else happens
EARLY	Relative to clock time

Guide word	Meaning (IEC 61882)
LATE	Relative to clock time
BEFORE	Relative to order or sequence
AFTER	Relative to order or sequence

Parameters used: flow, pressure, temperature, level, composition, heat transfer, measurement, control, integrity, maintenance, safety system and utility; for the procedure node also sequence, warm-up, venting, trap line and ESD. A deviation is recorded only when it is physically meaningful ("REVERSE temperature" is not).

3.4 Risk ranking

S \ L	L1	L2	L3	L4	L5
S5	5	10 (2)	15	20	25
S4	4 (2)	8 (7)	12 (3)	16	20
S3	3	6 (8)	9 (15)	12	15
S2	2	4 (4)	6 (12)	8 (5)	10
S1	1	2	3	4	5

Numbers in brackets: how many deviations of this study fall in the cell. Risk $R = S \times L$.

S	Severity (worst credible consequence)
1	First aid; minor damage; no release
2	Medical treatment; short loss of hot BFW; small contained release
3	Lost-time injury (scald); heater out of service for days; boiler-house upset
4	Permanent disability or single fatality; major exchanger damage; boiler trip or damage
5	Multiple fatalities; shell rupture with a large steam and hot-water release

L	Likelihood with existing safeguards
1	Remote: not expected in the life of the plant ($< 10^{-4}$ /y)
2	Unlikely: known in industry (10^{-4} - 10^{-3} /y)
3	Possible: could happen in the plant life (10^{-3} - 10^{-2} /y)
4	Likely: several times in the plant life (10^{-2} - 10^{-1} /y)
5	Frequent: yearly or more

1-4 Low acceptable · 5-9 Medium reduce if practicable (ALARP), review safeguards · 10-16 High recommendation required, consider LOPA · 20-25 Very high not acceptable, act before operation

4 HAZOP sessions

4.1 Schedule

The study is run in class as a teaching HAZOP. A typical plan is three sessions; the team records the dates and attendance on this page.

Session	Nodes	Date	Remark
1	N1 - N2 (BFW supply and pumps, flow control and tube side)		
2	N3 - N4 (steam supply, E-101 shell, vent, relief)		
3	N5 - N6 (condensate, procedures); review of recommendations		

4.2 Team

Role	Responsibility	Name	Attended sessions
HAZOP leader (chair)	Independent of the design; runs the method, keeps the team on the node, decides when a deviation is covered		
Scribe (recorder)	Writes the worksheet live, reads back recommendations		
Process engineer (designer)	Design intent, heat and mass balance, shell and tube design pressures, relief cases		
Operations representative	How the heater is really run: start-up venting and warm-up, trap checks, rounds, boiler-house communication		
Instrument / control engineer	Cascade TIC-101 → FIC-104, fail positions, SIS logic and trip independence		
Safety / risk engineer	Risk matrix, LOPA follow-up, regulations		
Specialists as needed	Mechanical (tube bundle, water hammer, vacuum rating), water treatment, boiler-house and upstream flash-plant representatives		

4.3 Documents reviewed

Document	Title / content
P&ID-001 rev C	E-101 P&ID, screen drawing of the simulator (Appendix 2)
BFWH-V1-PID-001 rev D	E-101 P&ID, alternative EPC-standard drawing (Appendix 2)
BFW-PFD-100	Process flow diagram with the stream table
BFWH-V1-IN-CE-001 rev R0	Cause and effect diagram of SIS I-100
BFWH-V1-IN-SRS-001 rev R0	Safety requirements specification of the SIFs
Simulator tabs	Process, Design, Control loop design, alarm and trip list (Workbook tab), start-up and shutdown procedures (SOP tab)

4.4 Nodes reviewed and guide-word coverage

Node	NO / NOT	MORE	LESS	AS WELL AS	PART OF	REVERSE	OTHER THAN	EARLY	LATE	BEFORE	AFTER	Rows	Flow	Pressure	Temperature / heat
N1	1	3	4	1	·	1	1	·	·	·	·	11	yes	yes	yes
N2	1	3	3	·	·	1	1	·	·	·	·	9	yes	yes	yes
N3	1	4	1	1	·	·	2	·	·	·	·	9	yes	yes	yes
N4	·	4	2	1	·	·	2	·	·	·	·	9	yes	yes	yes
N5	·	4	1	1	·	1	2	·	·	·	·	9	yes	yes	yes
N6	4	·	·	·	1	·	1	1	1	2	1	11	yes	yes	yes

A dot means the guide word gave no meaningful deviation for any parameter of the node (it was still considered). EARLY / LATE / BEFORE / AFTER apply mainly to the procedure node N6. The right-hand columns confirm that flow, pressure and temperature (or their equivalents) were examined in every process node.

5 Findings and recommendations

5.1 Risk profile

Node	Deviations	High	Medium	Low	Recommendations
N1 BFW supply: TK-101 → P-101A/B → recycle FV-101B	11	1	8	2	R01, R02, R03, R04, R07, R25
N2 BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	9	0	8	1	R04, R05, R06, R07, R25
N3 Steam supply: header → XV-100 → FT-104 → FV-104	9	1	7	1	R08, R09, R10, R24, R26
N4 E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	9	1	8	0	R08, R11, R12, R13, R14, R17, R27, R28
N5 Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	9	1	8	0	R12, R15, R16, R17, R28
N6 Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	11	1	8	2	R18, R19, R20, R21, R22
Total	58	5	47	6	28

High-ranked deviations

Ref	Deviation	Consequence	S×L	Rec.
N1.1	Flow NO: No BFW from the pumps	No BFW to the boiler house: boiler low drum level, boiler trip or dry firing (outside the unit). Steam on stagnant water in the E-101 tubes: local boiling and hammer. Shell drawn to vacuum once the steam is cut	4×3 = 12	R01, R02, R07
N3.3	Pressure MORE: Shell above its design pressure from the steam side	Shell pressure rises toward the header pressure; shell or nozzle rupture, large steam release, fatalities nearby. Model: PSHH-100 high shell pressure after 55 s, shell peak 256 kPa g	5×2 = 10	R08, R23
N4.2	Pressure MORE: Shell overpressure from a tube rupture	Shell fills with water and rises toward the BFW pressure: shell rupture, hot-water and flash-steam release	5×2 = 10	R11, R12, R23
N5.1	Level MORE: Condensate floods the shell (trap or strainer blocked)	Lower tubes flooded: outlet falls (40.4 °C after 30 min at 90 % blocking); water hammer when the level reaches about 70 % at 34:43: nozzle or shell damage, steam release, scalds	4×3 = 12	R12, R15
N6.2	Warm-up EARLY: Steam raised early, before the shell is warm	Water hammer: at 10 % output the model bangs 6.8 s after XV-100 opens in a cold shell (1.8 s at 40 %; none at 5 %); gasket or nozzle failure, scalds	4×3 = 12	R18

5.2 Recommendations

No.	Recommendation	Node	Raised from	Action by
R01	Auto-start P-101B on a P-101A trip or low discharge pressure (PI-106 transmitter, recommended addition), completing inside the FSLL-101 delay; test the spare start monthly.	N1	N1.1, N1.11, N6.11	Instrument / Operations
R02	Add level transmitter LT-101 (recommended addition) on TK-101 with a low-level alarm and a low-low pump trip, and a high-level alarm for overfill.	N1	N1.1, N1.2, N1.3, N1.4, N1.6	Instrument
R03	Provide a fixed minimum-flow line with restriction orifice from the pump discharge to TK-101, independent of FV-101B, so a blocked outlet downstream of E-101 cannot deadhead the running pump.	N1	N1.5	Process
R04	Confirm a check valve on each pump discharge and on the BFW line to the boiler house; add them to the inspection plan.	N1/N2	N1.7, N2.9	Mechanical
R05	Fit a thermal relief valve on the E-101 tube side (recommended addition), or lock the procedure so the steam is isolated and the shell cooled before the BFW is blocked in.	N2	N2.3, N2.4	Process
R06	Add a deviation alarm between TT-101 (control) and TIAH-108 (independent) so a transmitter reading low or frozen is seen before the boiler house is affected.	N2	N2.2, N2.5, N2.7	Instrument
R07	Agree the interface with the boiler house: shared alarms for FIC-101 LO and TIAH-108 HI, a call-out procedure, and include loss of BFW and hot BFW from E-101 in the boiler-house HAZOP.	N1/N2	N1.1, N1.2, N2.1, N2.2, N2.5	Operations / Process

No.	Recommendation	Node	Raised from	Action by
R08	Confirm the E-101 shell design pressure is at least the highest credible header pressure (144 °C saturation at 300 kPa g in the fault library) and that PSV-100 is set at or below it and sized for FV-104 fully open at that header pressure and for the external-fire case.	N3/N4	N3.1, N3.3, N4.5, N4.8	Process
R09	Header-pressure response procedure with the upstream flash-steam plant for PI-105 LO/HI; consider a pressure-control station on the header so E-101 never sees more than its design pressure.	N3	N3.2, N3.4	Operations / Process
R10	Add FV-104 position feedback ZT-104 (recommended addition) with a deviation alarm between FIC-104 output and valve position (stuck or failed valve).	N3	N3.1, N3.6, N3.7	Instrument
R11	Apply the tube-rupture (two-thirds) rule: BFW side up to 1030 kPa g at pump shut-off. If the shell design pressure is below two-thirds of the tube-side design pressure, size PSV-100 for the tube-rupture flow of water and flashing water.	N4	N4.2	Process
R12	Add a shell hotwell level transmitter LT-100 (recommended addition) with a high alarm below the tube bottom (12 % of the shell) and a high-high level trip closing XV-100 before steam can meet a flooded shell.	N4/N5	N4.2, N4.3, N5.1, N5.3	Instrument / Process
R13	Put a continuous vent in service (rev B RO-101) or the automatic air vent with vent-line temperature (rev C AAV-101, TI-102, TAL-102): gas blanketing gives no alarm for a long time.	N4	N2.6, N4.1, N6.3	Process
R14	Confirm the shell is rated for full vacuum or that VB-101 is sized for the fastest condensation after a trip; add VB-101 to the preventive-maintenance and test plan.	N4	N3.5, N4.4	Mechanical
R15	Trap management: regular trap survey (ultrasonic or temperature), Y-101 cleaning after any maintenance, and confirm ST-101 capacity at low shell differential pressure (start-up and near-vacuum operation).	N5	N2.6, N5.1, N5.3, N5.4, N5.5, N5.7	Mechanical / Operations
R16	Replace local gauge PG-110 by a return-header pressure transmitter with a high alarm (recommended addition); route shell drain V-105 to a flash pot or blowdown pit, not to an open sewer.	N5	N5.2, N5.6	Process / Instrument
R17	Route V-104, PSV-100, V-103 and V-105 discharges to a safe location; fit a silencer on the vent; warning signs and hearing-protection zone around the vent and trap set.	N4/N5	N4.7, N4.8, N5.6	Mechanical / Safety
R18	Start-up procedure hold points: BFW in AUTO first, vent V-104 open, warm-up with FIC-104 below 10 % (the model gives water hammer at 10 % in a cold shell, none at 5 %) until the shell is warm, vent until dry steam blows, then cascade.	N6	N6.2, N6.3, N6.7	Operations
R19	Shutdown procedure: steam off and shell depressured before the BFW is stopped or blocked in; open V-105 only after the shell pressure has fallen; keep V-104 open during lay-up.	N6	N2.4, N6.5, N6.6	Operations
R20	Trip-bypass management for FSLL-101, TSHH-108 and PSHH-100: permit, time limit, MOC, bypass status shown on the DCS; proof-test the XV-100 trips at the interval set by LOPA.	N6	N6.1, N6.9	Operations / Instrument
R21	Restart after any trip: re-vent the shell (air drawn in through VB-101) before the cascade is put back; consider "V-104 open" as a start permissive.	N6	N3.5, N6.8	Operations / Instrument
R22	Confirm the fail position of FV-101B (not shown on the P&ID) and that XV-100 closes on loss of air or power; document the fail positions in the cause-and-effect chart.	N6	N6.10, N6.11	Instrument
R23	LOPA for the High rows that rely on PSHH-100, FSLL-101 and PSV-100 (shell overpressure, tube rupture, loss of BFW, water hammer).	all	N3.3, N4.2	Process safety
R24	Show a calculated superheat TI-100 – Tsat(PI-100) on the DCS; tell the upstream plant when it exceeds a few degrees.	N3	N3.8	Instrument
R25	Water-treatment control: BFW hardness and oxygen-scavenger sampling programme; closed sampling point and PPE for treatment chemicals.	N1/N2	N1.10, N2.8	Operations
R26	Confirm a drip leg and trap on the steam branch upstream of XV-100, and warm the line with its drains open before admitting steam.	N3	N3.9	Mechanical / Operations
R27	Condensate quality: oil-in-condensate check or monitor on the return to the boiler house; find and stop the oil source in the flash steam.	N4	N4.6	Operations
R28	Maintenance isolation procedure for E-101: positive isolation (blinds) of steam, BFW and condensate, drain and cool before opening, confined-space permit for shell entry, lift plan for bundle pulls.	N4/N5	N4.9, N5.9	Operations / Maintenance

5.3 Dynamic check of selected deviations on the simulator

A conventional HAZOP judges consequences qualitatively. Here selected deviations are run on copies of the E-101 model from the design point with the existing safeguards active (alarms, SIS trips, controllers in their normal modes, no operator action) for up to 30 min, or 30 s past a trip. Alarms are evaluated from the page's alarm list every second, with the same suppression as the console (steam alarms only while steaming, BFW alarms only with a pump running). The runs show whether the safeguards act before harm, and how much time the operator has.

Node	Deviation	Initiating event (on the model)	First alarms (mm:ss)	Trip (first-out)	Range reached (to 30 s after a trip, or in 30 min)	Also seen
N1	Flow NO	Running pump P-101A trips; spare P-101B not started	none in 30 min	FSLL-101 low BFW flow (steam on stagnant water) at 00:07	FIC-101 0.0 to 100.0 m ³ /h TIAH-108 70.2 to 72.8 °C PI-100 –3 to 193 kPa g	FV-104 0–63 %
N2	Flow MORE	FV-101A fails open (recycle FV-101B closes)	PDI-109 HI at 00:04 FIC-101 HI at 00:05 TIC-101 LO at 00:16	no trip	FIC-101 100.0 to 125.6 m ³ /h TT-101 63.9 to 70.0 °C PDI-109 60 to 95 kPa	FV-104 47–100 %
N2	Flow LESS	FIC-101 forced to MAN at 25 % output	FIC-101 LO at 00:04 TIC-101 HI at 00:17	no trip	FIC-101 60.3 to 100.0 m ³ /h TIAH-108 68.7 to 82.9 °C PI-100 –3 to 164 kPa g	FV-104 20–64 % hotwell up to 17 % (tubes from 12 %)
N2	Temperature MORE	TT-101 reads 15 °C low: TIC-101 adds steam	TIC-101 LO at 00:13	no trip	TIAH-108 70.2 to 74.5 °C TT-101 57.1 to 70.0 °C PI-100 105 to 167 kPa g	FV-104 50–100 %
N3	Flow MORE	FV-104 steam valve fails open	FIC-104 HI at 00:02 TIC-101 HI at 00:46	no trip	PI-100 105 to 167 kPa g TIAH-108 70.2 to 74.5 °C FIC-104 9935 to 13432 kg/h	FV-104 50–100 %
N3	Pressure MORE	FV-104 fails open with the header rising toward 300 kPa g (two failures, checks PSHH-100)	FIC-104 HI at 00:02 PI-100 HI at 00:23 TIC-101 HI at 00:26	PSHH-100 high shell pressure at 00:55	PI-105 194 to 300 kPa g PI-100 –3 to 255 kPa g TIAH-108 56.9 to 78.1 °C	FV-104 50–100 % hotwell up to 7 % (tubes from 12 %)
N3	Pressure LESS	Steam header falls toward 115 kPa g	PI-105 LO at 00:33 TIC-101 LO at 01:03	no trip	PI-105 110 to 195 kPa g PI-100 59 to 105 kPa g TT-101 66.6 to 70.0 °C	FV-104 50–100 %
N4	Composition AS WELL AS	Non-condensable gas in the steam, vent V-104 closed	none in 30 min	no trip	PI-100 105 to 172 kPa g FIC-104 9927 to 10070 kg/h TT-101 69.9 to 70.0 °C	FV-104 50–100 %
N4	Level MORE	Tube leak into the shell (1.5 kg/s at full size)	none in 30 min	no trip	PI-100 81 to 119 kPa g FIC-104 9935 to 10841 kg/h TT-101 68.2 to 70.0 °C	FV-104 47–60 % hotwell up to 14 % (tubes from 12 %)
N5	Level MORE	Strainer Y-101 / trap ST-101 blocking (90 %)	TIC-101 LO at 08:36	no trip	PI-100 105 to 195 kPa g TT-101 40.4 to 70.0 °C FIC-104 4334 to 9935 kg/h	FV-104 50–100 % hotwell up to 67 % (tubes from 12 %)

5.4 Follow-up and close-out

- Each recommendation gets an owner and a due date; the owner answers *accepted and done*, *accepted with an alternative* (with justification) or *rejected* (with justification agreed by the HAZOP leader and the plant manager).
- High scenarios whose protection relies on instrumented trips or on PSV-100 (shell overpressure from the steam side, tube rupture, loss of BFW to the boiler house, water hammer) go to **LOPA** to set the required SIL of the XV-100 trips (IEC 61511) and to confirm the relief case.
- Any change (putting rev B RO-101 or rev C AAV-101 in service, a shell level transmitter, a new trip or a changed set point) goes through **management of change**, which checks whether the HAZOP must be updated.
- Interfaces: the loss-of-BFW and hot-BFW consequences are passed to the boiler-house HAZOP; header-pressure causes to the upstream flash-steam plant; trap and return-header causes to the condensate-system owner.
- Revalidate periodically (typically every five years) and after incidents.
- Use the simulator to train operators on the scenarios the HAZOP found, and to check that the procedures and alarm responses really work (section 9).

Appendix 1 · HAZOP worksheets

Node N1 · BFW supply: TK-101 → P-101A/B → recycle FV-101B

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Node	N1 · BFW supply: TK-101 → P-101A/B → recycle FV-101B
Design intent	Transfer treated BFW at 18 °C from TK-101 with P-101A (P-101B spare) at about 662 kPa g discharge; 10.7 m³/h returns to TK-101 through FV-101B so the running pump always has a minimum flow.	Design conditions	18 °C; discharge 662 kPa g; shut-off 1030 kPa g; 110.7 m³/h pumped
Equipment	TK-101, P-101A/B, suction and discharge piping, FV-101B recycle	Instruments	PI-106, TI-103, FIC-101 (flow seen by the pumps)
Drawing	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Document	BFWH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.1	Flow	NO	No BFW from the pumps	- Running pump P-101A trips (motor, power) - P-101B unavailable (motor fault) or not started - TK-101 empty - suction valve closed	No BFW to the boiler house: boiler low drum level, boiler trip or dry firing (outside the unit). Steam on stagnant water in the E-101 tubes: local boiling and hammer. Shell drawn to vacuum once the steam is cut	- FSLL-101 trip closes XV-100 (< 30 m³/h, 5 s) (model: trip at 7.5 s) - FIC-101 LO alarm (80 m³/h) - P-101B spare (manual start) - VB-101	4	3	12 H	R01, R02, R07	Instrument · Instrument / Operations · Operations / Process
N1.2	Flow	LESS	Less BFW from the pumps	- Pump wear or cavitation - low TK-101 level - partly closed suction valve	BFW short at the boiler house; hotter outlet at the same steam; pump damage	- FIC-101 LO alarm (80 m³/h) - TIC-101 cuts the steam - TI AH-108 HI alarm (85 °C)	3	3	9 M	R02, R07	Instrument · Operations / Process
N1.3	Level	LESS	Low level in TK-101	- Make-up to TK-101 stopped - tank drain left open	Pump cavitation, then no flow (see N1 Flow NO)	Operator rounds only (no level instrument on the P&ID)	3	3	9 M	R02	Instrument
N1.4	Level	MORE	High level in TK-101	- Make-up more than consumption - recycle high at low demand	Overflow of treated water with chemicals to drain	Tank overflow line (make-up system)	2	2	4 L	R02	Instrument
N1.5	Pressure	MORE	Pump deadheaded	Valve closed downstream of E-101 or at the boiler house while FIC-101 drives FV-101A open and FV-101B shut	Pump at shut-off (1030 kPa g) with no flow: water overheats, seal fails, hot-water spray	- FV-101B complementary recycle (protects only against FV-101A closing) - PI-106 local reading	3	2	6 M	R03	Process
N1.6	Pressure	LESS	Low suction pressure (cavitation)	- Low TK-101 level - hot BFW in TK-101 - plugged suction	Cavitation, noise, flow loss, pump damage	- TI-103 HI alarm (25 °C) - FIC-101 LO alarm (80 m³/h)	2	3	6 M	R02	Instrument
N1.7	Flow	REVERSE	Reverse flow through the idle pump	Passing check valve on the idle pump	Recirculation through the idle pump, less forward flow, idle pump spins backwards	- Check valves (assumed, not shown) - FIC-101 LO alarm (80 m³/h)	2	2	4 L	R04	Mechanical
N1.8	Temperature	MORE	Hot BFW in TK-101	- Hot make-up or condensate routed to the tank - BFW inlet up to 15 °C hotter	Less NPSH; outlet control keeps 70 °C with less steam	- TI-103 HI alarm (25 °C) - TIC-101 cuts the steam	2	3	6 M	—	—
N1.9	Temperature	LESS	Cold BFW (cold week)	Cold make-up water	More steam needed (14.6 % more in the model, shell 153 kPa g): FV-104 loses its margin; outlet may fall below target	- TI-103 LO alarm (12 °C) - TIC-101 adds steam - TIC-101 LO alarm (67 °C)	2	3	6 M	—	—
N1.10	Composition	AS WELL AS	Oxygen, hardness or wrong treatment chemical in the BFW	- Dosing upset - poor make-up quality	Corrosion or scale in the tubes (tube leak later), boiler deposits; exposure when sampling	Water-quality sampling programme	2	3	6 M	R25	Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N1.11	Maintenance	OTHER THAN	Pump change-over done wrongly	- Spare started against a closed discharge - running pump stopped first	BFW interrupted: FSLL-101 closes XV-100; boiler house loses BFW	- Change-over procedure - FSLL-101 trip closes XV-100 (< 30 m³/h, 5 s)	3	3	9 M	R01	Instrument / Operations

Node N2 · BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Node	N2 · BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house
Design intent	Deliver 100 m ³ /h of BFW through the 100 tubes of E-101 (FV-101A, fail open), heat it from 18 to 70 °C and send it to the boiler house deaerator at about 602 kPa g.	Design conditions	100 m ³ /h; 18 → 70 °C; tube ΔP 60 kPa; outlet 602 kPa g
Equipment	FV-101A, FT-101, E-101 tubes (one pass), outlet line to the deaerator	Instruments	FIC-101, FSLL-101, TI-103, PDI-109, TT-101 / TIC-101, TIAH-108 / TSHH-108
Drawing	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Document	BFWH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.1	Flow	MORE	More BFW through the tubes	- FV-101A fails open (fail-open design) - boiler house raises the set point - FT-101 reads low	Pump runs out on its curve (125.6 m ³ /h in the model), recycle closes, outlet falls below target, boiler house gets cooler water	- FIC-101 HI alarm (120 m³/h) - PDI-109 HI alarm (80 kPa) - TIC-101 LO alarm (67 °C)	2	3	6 M	R07	Operations / Process
N2.2	Flow	LESS	Less BFW through the tubes	- FIC-101 output low or forced to MAN - FV-101A sticks - FT-101 reads high	Outlet hotter: with 60 m ³ /h and FV-104 wide open the model reaches 85.5 °C; below the FSLL setting, steam on nearly stagnant water boils in the tubes	- FIC-101 LO alarm (80 m³/h) - FSLL-101 trip closes XV-100 (< 30 m³/h, 5 s) - TIAH-108 HI alarm (85 °C) - TSHH-108 trip closes XV-100 (> 95 °C, 5 s)	3	3	9 M	R06, R07	Instrument · Operations / Process
N2.3	Flow	NO	No flow through the tubes with steam on	- FV-101A closed by a controller fault - outlet valve closed at the boiler house	Water boils in the tubes: pressure surges and hammer on the tube side; blocked-in water expands	- FSLL-101 trip closes XV-100 (< 30 m³/h, 5 s) - FV-101A fails open	4	2	8 M	R05	Process
N2.4	Pressure	MORE	Tube side blocked in while hot	Inlet and outlet isolated for maintenance with steam still on or the shell still at 121.3 °C	Liquid thermal expansion overpressures the tube side: gasket or flange leak of hot water, tube damage	None on the P&ID (no tube-side relief)	4	2	8 M	R05, R19	Operations · Process
N2.5	Temperature	MORE	BFW to the boiler house hotter than intended	- TT-101 reads low - TIC-101 or FIC-104 in MAN at high output - FV-104 fails open - low BFW flow	Hot BFW to the deaerator: flashing, deaerator pressure rise, boiler feed pump cavitation. The model reaches 74.3 °C at the design header and 85.5 °C at 60 m ³ /h, below the TSHH setting	- TIAH-108 HI alarm (85 °C) - TSHH-108 trip closes XV-100 (> 95 °C, 5 s) - TIC-101 HI alarm (74 °C)	3	2	6 M	R06, R07	Instrument · Operations / Process
N2.6	Temperature	LESS	BFW to the boiler house colder than intended	- Low header pressure - gas blanketing - condensate flooding - tube scale - FV-104 stuck	Boiler house needs more heat; deaerator steam demand up (operability)	TIC-101 LO alarm (67 °C)	2	4	8 M	R13, R15	Mechanical / Operations · Process
N2.7	Measurement	OTHER THAN	TT-101 wrong or frozen	Transmitter drift low, frozen reading, wiring fault	The cascade acts on a wrong value: over- or under-heating	- TIAH-108 independent measurement - local gauge TG-101	3	3	9 M	R06	Instrument
N2.8	Heat transfer	LESS	Scale inside the tubes	- Hardness in the BFW - poor treatment	Duty lost, FV-104 wide open, outlet falls (54.8 °C at full scale in the model); higher tube ΔP	- PDI-109 HI alarm (80 kPa) - TIC-101 LO alarm (67 °C)	2	3	6 M	R25	Operations

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N2.9	Flow	REVERSE	Back-flow from the boiler house	• Pump stops while the deaerator side is at higher pressure • no check valve	Hot deaerator water back through E-101 to TK-101; pump turns backwards	None shown	2	2	4 L	R04	Mechanical

Node N3 · Steam supply: header → XV-100 → FT-104 → FV-104

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Node	N3 · Steam supply: header → XV-100 → FT-104 → FV-104
Design intent	Take 9,935 kg/h of flash steam from the header at about 195 kPa g and throttle it with FV-104 (fail closed) under FIC-104, whose set point comes from TIC-101; XV-100 isolates the steam on any trip.	Design conditions	Header 200 kPa g (133.9 °C sat.); 9,935 kg/h; FV-104 50 % open
Equipment	Steam branch, XV-100 (SIS), FT-104, FV-104	Instruments	PI-105, FIC-104, TIC-101 (cascade master), I-100 (SIS)
Drawing	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Document	BFWH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.1	Flow	MORE	More steam to the shell	- FV-104 fails open - TIC-101 or FIC-104 forced to MAN at high output - FT-104 reads low	Shell pressure rises to about 167 kPa g and the outlet to 74.3 °C at the design header; hot BFW; if the header is also high, see the next row	- TIC-101 HI alarm (74 °C) - FIC-104 HI alarm (13,000 kg/h) - PI-100 HI alarm (200 kPa g) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100	3	3	9 M	R08, R10	Instrument · Process
N3.2	Pressure	MORE	Header pressure high	- Upstream flash-plant upset (up to 300 kPa g) - upstream let-down failure	FV-104 throttles to about 30 %; the shell briefly reaches 138 kPa g; control margin at the low end is lost	- PI-105 HI alarm (240 kPa g) - FIC-104 flow control - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100	3	3	9 M	R09	Operations / Process
N3.3	Pressure	MORE	Shell above its design pressure from the steam side	FV-104 fails open, or the cascade is forced high, while the header is at its highest	Shell pressure rises toward the header pressure; shell or nozzle rupture, large steam release, fatalities nearby. Model: PSHH-100 high shell pressure after 55 s, shell peak 256 kPa g	- PI-100 HI alarm (200 kPa g) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100 (set pressure not on the P&ID) - XV-100 closes on trip	5	2	10 H	R08, R23	Process · Process safety
N3.4	Pressure	LESS	Header pressure low	Upstream plant at reduced rate (toward 115 kPa g)	FV-104 wide open; shell 82 kPa g; outlet 68.1 °C, below target (operability)	- PI-105 LO alarm (150 kPa g) - TIC-101 LO alarm (67 °C)	2	4	8 M	R09	Operations / Process
N3.5	Flow	NO	No steam	- XV-100 closed (trip, spurious trip, loss of air or power) - FV-104 fails closed - header lost	Cold BFW to the boiler house; shell drawn to vacuum (–3 kPa g in the model) and air enters through VB-101; restart without venting leaves the shell blanketed	- VB-101 - TIC-101 LO alarm (67 °C) - restart procedure	3	3	9 M	R14, R21	Mechanical · Operations / Instrument
N3.6	Control	OTHER THAN	FV-104 stuck	- Actuator or positioner failure - packing friction	Steam does not follow demand: outlet 65.2 °C with the BFW at 120 m³/h in the model; overheating if the BFW falls instead	- TIC-101 LO alarm (67 °C) - TIC-101 HI alarm (74 °C) - operator compares FIC-104 output with the flow	2	3	6 M	R10	Instrument
N3.7	Measurement	OTHER THAN	FT-104 reads low	Transmitter or impulse-line fault	FIC-104 opens FV-104; transient overheating (outlet up to 74.3 °C) until TIC-101 corrects; steam accounting wrong	- TIC-101 HI alarm (74 °C) - TI-108 HI alarm (85 °C)	2	3	6 M	R10	Instrument
N3.8	Temperature	MORE	Superheated steam	Let-down upstream without desuperheating (up to 40 °C superheat)	Hotter inlet nozzle and surfaces; a dry-wall zone at the inlet; shell runs higher (143 kPa g in the model)	TI-100 read against the saturation temperature of PI-100	2	2	4 L	R24	Instrument

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N3.9	Composition	AS WELL AS	Condensate slugs in the steam branch	• No drip leg upstream of XV-100 • cold line at start-up	Water hammer in the steam line: damage to FV-104, FT-104, flanges; steam release	None shown (line drainage upstream not on the P&ID)	4	2	8 M	R26	Mechanical / Operations

Node N4 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Node	N4 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100
Design intent	Condense 9,935 kg/h of steam at about 105 kPa g (121.3 °C) on the whole tube surface, keep non-condensable gas vented, and keep the shell between atmospheric pressure and its design pressure.	Design conditions	105 kPa g; 121.3 °C; duty 6.04 MW; hotwell 5 % (tubes from 12 %)
Equipment	E-101 shell, V-104 air vent (rev B RO-101, rev C AAV-101), VB-101, PSV-100	Instruments	PI-100, PSHH-100, TI-100 (TI-102 / TAL-102 with rev C)
Drawing	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Document	BFWH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N4.1	Composition	AS WELL AS	Non-condensable gas in the shell	- Gas carried in the flash steam - air drawn in through VB-101 not vented at restart - vent closed	Blanketed surface: shell pressure creeps up (172 kPa g after 30 min) with no alarm; outlet 65.3 °C after 1 h and 56.7 °C after 2 h; CO ₂ gives corrosive condensate	- V-104 manual vent - TIC-101 LO alarm (67 °C) (late) - rev B / rev C not in service	2	4	8 M	R13	Process
N4.2	Pressure	MORE	Shell overpressure from a tube rupture	- Tube rupture (corrosion, vibration, fatigue, water hammer) - BFW side at 662 kPa g (1030 kPa g at shut-off)	Shell fills with water and rises toward the BFW pressure: shell rupture, hot-water and flash-steam release	- PSV-100 (tube-rupture case not confirmed) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) stops the steam but not the BFW	5	2	10 H	R11, R12, R23	Instrument / Process · Process · Process safety
N4.3	Level	MORE	Tube leak into the shell	Pitting, rolled-joint leak	More condensate than steam; hotwell rises to 14.3 % (tubes start at 12 %); steam 9.1 % higher; BFW short at the boiler house; no alarm	- None automatic - condensate bucket test - steam-to-duty comparison	3	3	9 M	R12	Instrument / Process
N4.4	Pressure	LESS	Shell under vacuum	Steam cut (trip, ESD, FV-104 closed) while cold BFW flows	Shell falls to –3 kPa g within seconds: shell buckling if VB-101 fails; trap cannot discharge; air enters	VB-101	3	2	6 M	R14	Mechanical
N4.5	Temperature	MORE	External fire on the shell	Fire nearby (oil spill, adjacent unit)	Isolated shell heated: pressure rises, shell rupture	- PSV-100 (fire case to be confirmed) - fire water	4	2	8 M	R08	Process
N4.6	Heat transfer	LESS	Oil film on the tubes (shell side)	Oil carried in the flash steam from upstream	Duty falls (54.8 °C at full severity), PDI-109 unchanged; oil in the condensate to the boiler house	- TIC-101 LO alarm (67 °C) - condensate appearance	2	3	6 M	R27	Operations
N4.7	Flow	MORE	Vent V-104 left wide open	Not closed after start-up	Live steam lost (9.1 % more steam in the model); noise and scalding plume at the discharge	- Procedure (close to 10 % or less) - rev B / rev C	2	3	6 M	R17	Mechanical / Safety
N4.8	Integrity	OTHER THAN	PSV-100 lifts or passes	- Set too low - seat damage	Steam blowing to atmosphere, noise, scalds at the discharge	Discharge to a safe location	3	2	6 M	R08, R17	Mechanical / Safety · Process
N4.9	Maintenance	OTHER THAN	Shell entry or bundle pull	Inspection, tube plugging	Scalds from trapped steam or hot water, oxygen-deficient or hot atmosphere in the shell, dropped bundle	- Permit to work - isolation of steam, BFW and condensate - lifting procedure	4	2	8 M	R28	Operations / Maintenance

Node N5 · Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Node	N5 · Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain
Design intent	Remove 9,935 kg/h of condensate at about 119 °C to the return header at 30 kPa g, keep the hotwell at 5 % of the shell (below the tubes at 12 %) and hold steam in the shell.	Design conditions	119 °C; 30 kPa g return (107.4 °C saturation); hotwell 5 %
Equipment	V-101, Y-101 strainer, ST-101 trap, V-102, V-103 bypass, V-105 drain to sewer	Instruments	None on the P&ID (local PG-110 return pressure, TG-107 condensate temperature)
Drawing	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Document	BFWH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N5.1	Level	MORE	Condensate floods the shell (trap or strainer blocked)	- Y-101 screen plugged (scale, debris after maintenance) - trap stuck closed	Lower tubes flooded: outlet falls (40.4 °C after 30 min at 90 % blocking); water hammer when the level reaches about 70 % at 34:43: nozzle or shell damage, steam release, scalds	- TIC-101 LO alarm (67 °C) only - E-101 HAMMER event alarm (after the fact) - no level instrument	4	3	12 H	R12, R15	Instrument / Process · Mechanical / Operations
N5.2	Pressure	MORE	Return header back-pressure high	- Other users or flash in the common header - header valve partly closed	Trap discharges only when the shell is above the header: shell floods (level 39 %), outlet 59.9 °C, shell 182 kPa g; hammer risk	- Local gauge PG-110 - TIC-101 LO alarm (67 °C)	3	3	9 M	R16	Process / Instrument
N5.3	Pressure	LESS	Shell pressure below the return header	- Low steam demand (warm BFW, low BFW flow) or low header pressure throttles the shell - shell under vacuum after a trip	Trap has no driving pressure: condensate backs up and floods the lower tubes (the N2 low-flow run on the simulator floods the hotwell), hammer risk when the load returns	- TIC-101 LO alarm (67 °C) (late) - no level instrument	3	3	9 M	R12, R15	Instrument / Process · Mechanical / Operations
N5.4	Flow	MORE	Trap passes live steam (fails open)	Worn trap valve or seat	Steam 18.1 % higher for the same duty; return-header pressure up, hammer in the return line; scalds at leaks	- None automatic - trap surveys	3	3	9 M	R15	Mechanical / Operations
N5.5	Flow	AS WELL AS	Bypass V-103 left open	Left open after a trap problem	Live steam lost (7.7 % more steam in the model); flashing in the return header	- Procedure - trap-set rounds	2	3	6 M	R15	Mechanical / Operations
N5.6	Flow	OTHER THAN	Hot condensate drained to sewer	V-105 opened with the shell under pressure	Condensate at 119 °C flashes in the sewer: steam cloud, scalds, damage to the sewer, hot discharge (3.3 % more steam while open)	Procedure: drain only after the shell pressure has fallen	3	3	9 M	R16, R17	Mechanical / Safety · Process / Instrument
N5.7	Flow	REVERSE	Back-flow from the return header	Shell under vacuum after a trip while the header is at 30 kPa g	Condensate from the header back into the shell: flooding, hammer at restart	Trap resists reverse flow (check valve not shown)	3	2	6 M	R15	Mechanical / Operations
N5.8	Temperature	MORE	Condensate flashing in the return line	Normal: 119 °C condensate enters a 30 kPa g header (saturation 107.4 °C)	Two-phase flow and hammer in the return line	Line design (assumed adequate)	2	3	6 M	—	—
N5.9	Maintenance	OTHER THAN	Opening the strainer or trap while hot	- Block valves V-101 / V-102 passing - line not drained	Scalds from hot condensate and flash steam	- Permit to work - isolation and drain	3	3	9 M	R28	Operations / Maintenance

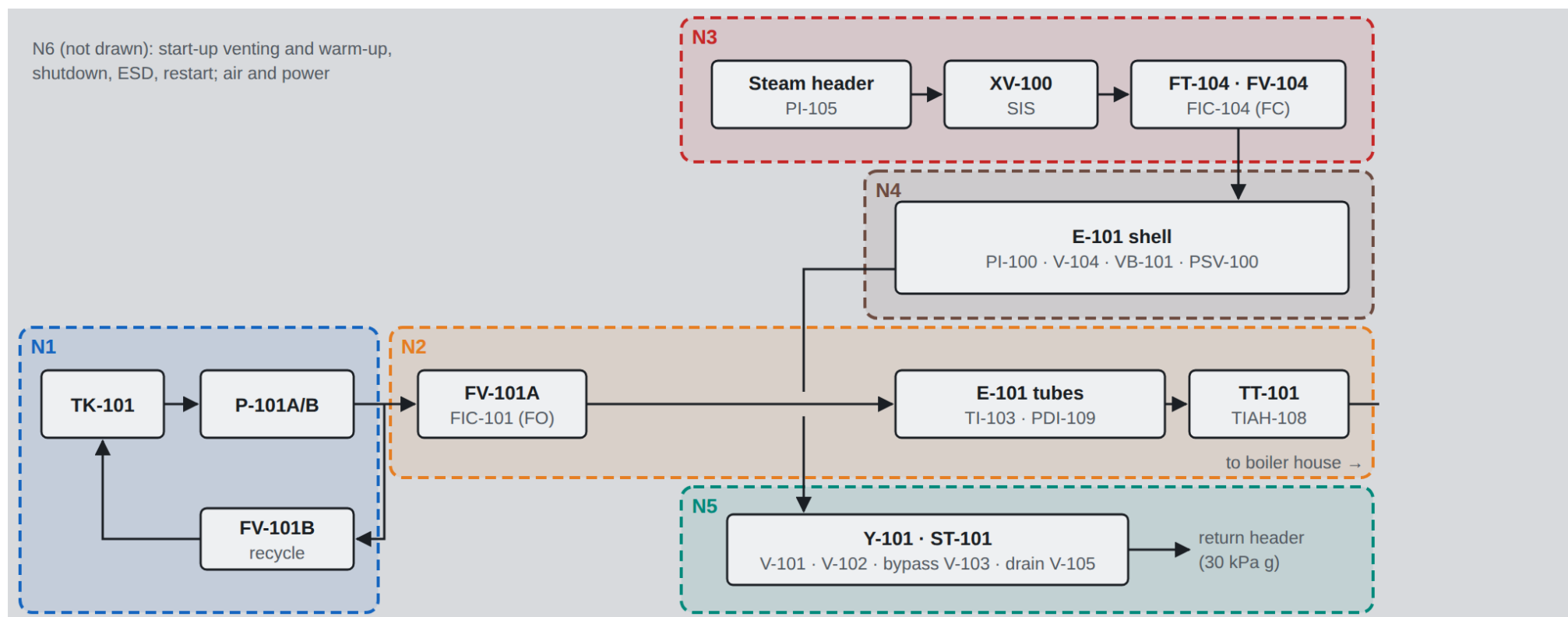
Node N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Node	N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities
Design intent	Bring E-101 from cold to 70 °C without water hammer or air blanketing, shut it down and restart it safely, and survive loss of air or power.	Design conditions	Start permissives: SIS reset, BFW $\geq 50 \text{ m}^3/\text{h}$, FIC-104 in MAN $\leq 10 \%$, trap line open
Equipment	Whole unit; SIS start permissives and trips	Instruments	SIS screen (XV-100), FIC-104 in MAN, V-104, V-105, V-101 / V-102
Drawing	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Document	BFWH-V1-SA-HZP-001 rev A

ID	Parameter	Guide word	Deviation	Causes	Consequences	Existing safeguards	S	L	Risk	Rec.	Action party
N6.1	Sequence	BEFORE	Steam admitted before the BFW flows	- Operator error - permissive bypassed	Steam on stagnant water: boiling in the tubes, overpressure, hammer	- XV-100 start permissive BFW $\geq 50 \text{ m}^3/\text{h}$ - FSLL-101 trip closes XV-100 ($< 30 \text{ m}^3/\text{h}$, 5 s)	4	1	4 L	R20	Operations / Instrument
N6.2	Warm-up	EARLY	Steam raised early, before the shell is warm	- FIC-104 opened too far at start-up - the permissive checks FIC-104 only at the moment XV-100 opens	Water hammer: at 10 % output the model bangs 6.8 s after XV-100 opens in a cold shell (1.8 s at 40 %; none at 5 %); gasket or nozzle failure, scalds	- Start permissive FIC-104 in MAN $\leq 10 \%$ - warm-up step in the procedure - E-101 HAMMER alarm	4	3	12 H	R18	Operations
N6.3	Venting	PART OF	Vent closed before dry steam blows	V-104 closed too early at start-up	Air left in the shell: surface blanketed, FV-104 wide open, cannot reach 70 °C	- Procedure step "vent until dry steam blows" - TIC-101 LO alarm (67 °C)	2	4	8 M	R13, R18	Operations · Process
N6.4	Trap line	NO	Trap line not in service at start-up	V-101 / V-102 left closed after maintenance	Condensate cannot leave: flooding and hammer	Start permissive: V-101 and V-102 open	4	1	4 L	—	—
N6.5	Sequence	AFTER	Steam isolated after the BFW is stopped	Pump stopped first at shutdown	Steam on stagnant water: boiling in the tubes	- FSLL-101 trip closes XV-100 ($< 30 \text{ m}^3/\text{h}$, 5 s) - shutdown procedure	3	2	6 M	R19	Operations
N6.6	Sequence	BEFORE	Drain opened before the shell is depressured	Hurry at shutdown	Flashing condensate at the drain: scalds	Procedure: open V-105 once the shell pressure has fallen	3	2	6 M	R19	Operations
N6.7	ESD	LATE	Steam not stopped quickly on an emergency order	Operator unfamiliar with the SIS screen	Hot BFW or steam keeps flowing during a boiler-house emergency	- ESD pushbutton on the SIS screen - procedure	3	2	6 M	R18	Operations
N6.8	Venting	NO	Restart without venting	SIS reset and steam readmitted with V-104 closed after a trip (air drawn in through VB-101)	Blanketed shell, low duty, cold BFW to the boiler house	Restart procedure	2	4	8 M	R21	Operations / Instrument
N6.9	Safety system	OTHER THAN	Trip bypassed or overridden	Testing, nuisance trips	No protection against low flow, high outlet temperature or high shell pressure	Permit to work	4	2	8 M	R20	Operations / Instrument
N6.10	Utility	NO	Loss of instrument air	Compressor trip, air-line break	FV-104 fails closed and XV-100 closes (steam off, shell to vacuum); FV-101A fails open (pump runs out); FV-101B position not stated	- Fail positions: FV-104 FC, FV-101A FO - VB-101	3	3	9 M	R22	Instrument
N6.11	Utility	NO	Loss of power	Grid or substation failure	Pumps stop: FSLL-101 closes XV-100; no BFW to the boiler house	- FSLL-101 trip closes XV-100 ($< 30 \text{ m}^3/\text{h}$, 5 s) - P-101B (on the same supply)	4	2	8 M	R01, R22	Instrument · Instrument / Operations

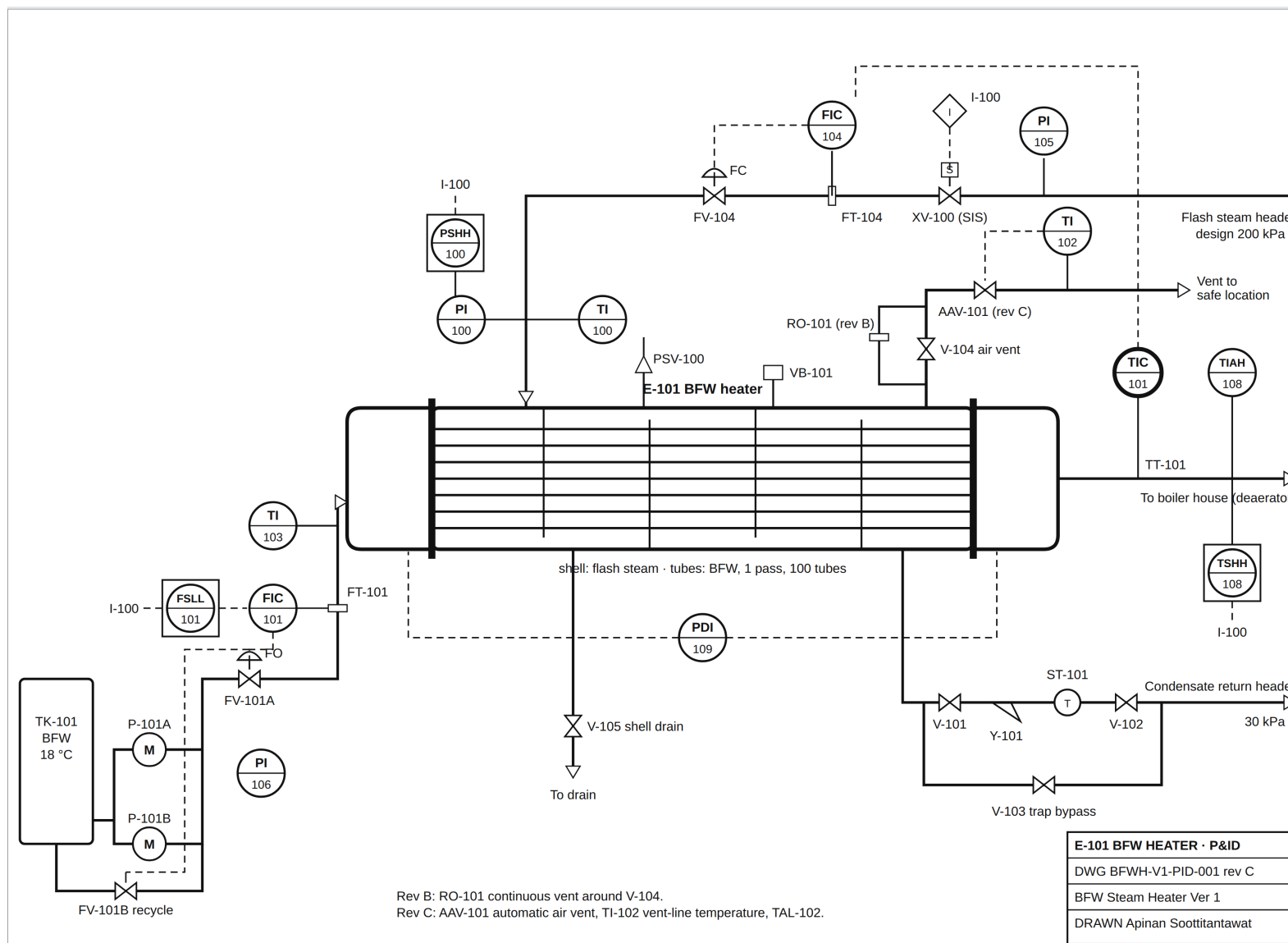
Appendix 2 · HAZOP documents (drawings with node boundaries)

A2-1 · HAZOP node map



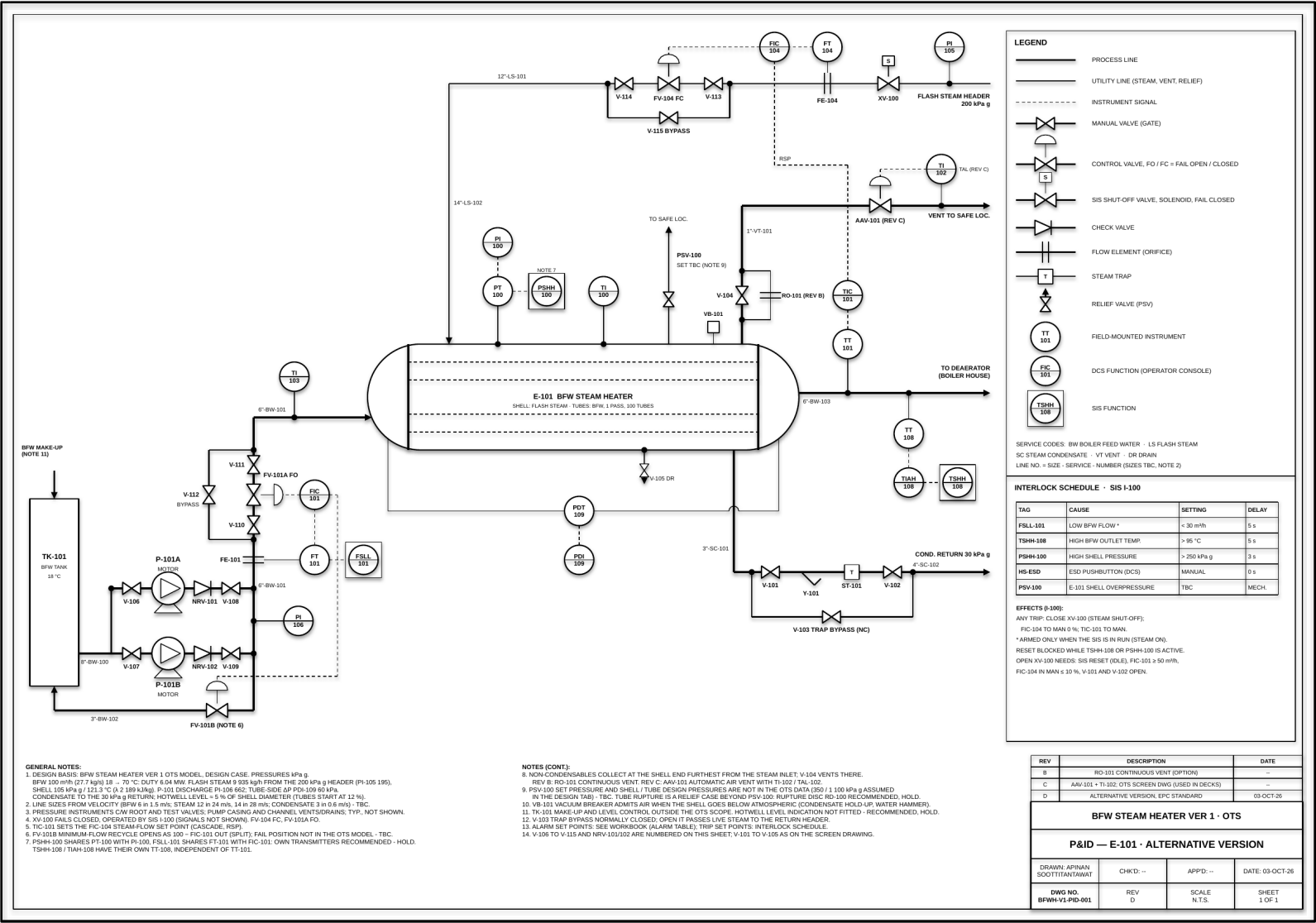
A2-1. HAZOP node map of E-101 (nodes N1–N5; N6 = procedures).

A2-2 · P&ID-001 rev C (screen drawing used for the study)



P&ID of E-101, rev C, as drawn on the P&ID tab of the simulator.

A2-3 · BFWH-V1-PID-001 rev D (alternative EPC drawing)



P&ID of E-101, rev D, EPC-standard alternative version with interlock schedule.

Appendix 3 · HAZOP action close-out sheets

One sheet per recommendation (28 sheets). The action party writes the response, ticks the status and signs; the HAZOP leader accepts the close-out.

HAZOP ACTION CLOSE-OUT SHEET · R01

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R01
Node	N1 · BFW supply: TK-101 → P-101A/B → recycle FV-101B	Action party	Instrument / Operations
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No BFW from the pumps	- Running pump P-101A trips (motor, power) - P-101B unavailable (motor fault) or not started - TK-101 empty - suction valve closed	No BFW to the boiler house: boiler low drum level, boiler trip or dry firing (outside the unit). Steam on stagnant water in the E-101 tubes: local boiling and hammer. Shell drawn to vacuum once the steam is cut	- FSL-101 trip closes XV-100 (< 30 m³/h, 5 s) (model: trip at 7.5 s) - FIC-101 LO alarm (80 m³/h) - P-101B spare (manual start) - VB-101	S4 L3 12 High
N1.11	Maintenance OTHER THAN : Pump change-over done wrongly	- Spare started against a closed discharge - running pump stopped first	BFW interrupted: FSL-101 closes XV-100; boiler house loses BFW	- Change-over procedure - FSL-101 trip closes XV-100 (< 30 m³/h, 5 s)	S3 L3 9 Medium
N6.11	Utility NO : Loss of power	Grid or substation failure	Pumps stop: FSL-101 closes XV-100; no BFW to the boiler house	- FSL-101 trip closes XV-100 (< 30 m³/h, 5 s) - P-101B (on the same supply)	S4 L2 8 Medium

HAZOP recommendation

Auto-start P-101B on a P-101A trip or low discharge pressure (PI-106 transmitter, recommended addition), completing inside the FSL-101 delay; test the spare start monthly.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R02

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R02
Node	N1 · BFW supply: TK-101 → P-101A/B → recycle FV-101B	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No BFW from the pumps	- Running pump P-101A trips (motor, power) - P-101B unavailable (motor fault) or not started - TK-101 empty - suction valve closed	No BFW to the boiler house: boiler low drum level, boiler trip or dry firing (outside the unit). Steam on stagnant water in the E-101 tubes: local boiling and hammer. Shell drawn to vacuum once the steam is cut	- FSL-101 trip closes XV-100 (< 30 m³/h, 5 s) (model: trip at 7.5 s) - FIC-101 LO alarm (80 m³/h) - P-101B spare (manual start) - VB-101	S4 L3 12 High
N1.2	Flow LESS : Less BFW from the pumps	- Pump wear or cavitation - low TK-101 level - partly closed suction valve	BFW short at the boiler house; hotter outlet at the same steam; pump damage	- FIC-101 LO alarm (80 m³/h) - TIC-101 cuts the steam - TIAH-108 HI alarm (85 °C)	S3 L3 9 Medium
N1.3	Level LESS : Low level in TK-101	- Make-up to TK-101 stopped - tank drain left open	Pump cavitation, then no flow (see N1 Flow NO)	Operator rounds only (no level instrument on the P&ID)	S3 L3 9 Medium
N1.4	Level MORE : High level in TK-101	- Make-up more than consumption - recycle high at low demand	Overflow of treated water with chemicals to drain	Tank overflow line (make-up system)	S2 L2 4 Low
N1.6	Pressure LESS : Low suction pressure (cavitation)	- Low TK-101 level - hot BFW in TK-101 - plugged suction	Cavitation, noise, flow loss, pump damage	- TI-103 HI alarm (25 °C) - FIC-101 LO alarm (80 m³/h)	S2 L3 6 Medium

HAZOP recommendation

Add level transmitter LT-101 (recommended addition) on TK-101 with a low-level alarm and a low-low pump trip, and a high-level alarm for overflow.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R03

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R03
Node	N1 · BFW supply: TK-101 → P-101A/B → recycle FV-101B	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.5	Pressure MORE : Pump deadheaded	Valve closed downstream of E-101 or at the boiler house while FIC-101 drives FV-101A open and FV-101B shut	Pump at shut-off (1030 kPa g) with no flow: water overheats, seal fails, hot-water spray	- FV-101B complementary recycle (protects only against FV-101A closing) - PI-106 local reading	S3 L2 6 Medium

HAZOP recommendation

Provide a fixed minimum-flow line with restriction orifice from the pump discharge to TK-101, independent of FV-101B, so a blocked outlet downstream of E-101 cannot deadhead the running pump.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R04

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R04
Node	N1/N2 · BFW supply: TK-101 → P-101A/B → recycle FV-101B / BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	Action party	Mechanical
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.7	Flow REVERSE : Reverse flow through the idle pump	Passing check valve on the idle pump	Recirculation through the idle pump, less forward flow, idle pump spins backwards	- Check valves (assumed, not shown) - FIC-101 LO alarm (80 m³/h)	S2 L2 4 Low
N2.9	Flow REVERSE : Back-flow from the boiler house	- Pump stops while the deaerator side is at higher pressure - no check valve	Hot deaerator water back through E-101 to TK-101; pump turns backwards	None shown	S2 L2 4 Low

HAZOP recommendation

Confirm a check valve on each pump discharge and on the BFW line to the boiler house; add them to the inspection plan.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R05

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R05
Node	N2 · BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.3	Flow NO : No flow through the tubes with steam on	- FV-101A closed by a controller fault - outlet valve closed at the boiler house	Water boils in the tubes: pressure surges and hammer on the tube side; blocked-in water expands	- FSL-101 trip closes XV-100 (< 30 m³/h, 5 s) - FV-101A fails open	S4 L2 8 Medium
N2.4	Pressure MORE : Tube side blocked in while hot	Inlet and outlet isolated for maintenance with steam still on or the shell still at 121.3 °C	Liquid thermal expansion overpressures the tube side: gasket or flange leak of hot water, tube damage	None on the P&ID (no tube-side relief)	S4 L2 8 Medium

HAZOP recommendation

Fit a thermal relief valve on the E-101 tube side (recommended addition), or lock the procedure so the steam is isolated and the shell cooled before the BFW is blocked in.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R06

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R06
Node	N2 · BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.2	Flow LESS : Less BFW through the tubes	- FIC-101 output low or forced to MAN - FV-101A sticks - FT-101 reads high	Outlet hotter: with 60 m ³ /h and FV-104 wide open the model reaches 85.5 °C; below the FSLL setting, steam on nearly stagnant water boils in the tubes	- FIC-101 LO alarm (80 m³/h) - FSLL-101 trip closes XV-100 (< 30 m³/h, 5 s) - TIAH-108 HI alarm (85 °C) - TSHH-108 trip closes XV-100 (> 95 °C, 5 s)	S3 L3 9 Medium
N2.5	Temperature MORE : BFW to the boiler house hotter than intended	- TT-101 reads low - TIC-101 or FIC-104 in MAN at high output - FV-104 fails open - low BFW flow	Hot BFW to the deaerator: flashing, deaerator pressure rise, boiler feed pump cavitation. The model reaches 74.3 °C at the design header and 85.5 °C at 60 m ³ /h, below the TSHH setting	- TIAH-108 HI alarm (85 °C) - TSHH-108 trip closes XV-100 (> 95 °C, 5 s) - TIC-101 HI alarm (74 °C)	S3 L2 6 Medium
N2.7	Measurement OTHER THAN : TT-101 wrong or frozen	Transmitter drift low, frozen reading, wiring fault	The cascade acts on a wrong value: over- or under-heating	- TIAH-108 independent measurement - local gauge TG-101	S3 L3 9 Medium

HAZOP recommendation

Add a deviation alarm between TT-101 (control) and TIAH-108 (independent) so a transmitter reading low or frozen is seen before the boiler house is affected.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R07

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R07
Node	N1/N2 · BFW supply: TK-101 → P-101A/B → recycle FV-101B / BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	Action party	Operations / Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.1	Flow NO : No BFW from the pumps	- Running pump P-101A trips (motor, power) - P-101B unavailable (motor fault) or not started - TK-101 empty - suction valve closed	No BFW to the boiler house: boiler low drum level, boiler trip or dry firing (outside the unit). Steam on stagnant water in the E-101 tubes: local boiling and hammer. Shell drawn to vacuum once the steam is cut	- FSL-101 trip closes XV-100 (< 30 m³/h, 5 s) (model: trip at 7.5 s) - FIC-101 LO alarm (80 m³/h) - P-101B spare (manual start) - VB-101	S4 L3 12 High
N1.2	Flow LESS : Less BFW from the pumps	- Pump wear or cavitation - low TK-101 level - partly closed suction valve	BFW short at the boiler house; hotter outlet at the same steam; pump damage	- FIC-101 LO alarm (80 m³/h) - TIC-101 cuts the steam - TIAH-108 HI alarm (85 °C)	S3 L3 9 Medium
N2.1	Flow MORE : More BFW through the tubes	- FV-101A fails open (fail-open design) - boiler house raises the set point - FT-101 reads low	Pump runs out on its curve (125.6 m ³ /h in the model), recycle closes, outlet falls below target, boiler house gets cooler water	- FIC-101 HI alarm (120 m³/h) - PDI-109 HI alarm (80 kPa) - TIC-101 LO alarm (67 °C)	S2 L3 6 Medium
N2.2	Flow LESS : Less BFW through the tubes	- FIC-101 output low or forced to MAN - FV-101A sticks - FT-101 reads high	Outlet hotter: with 60 m ³ /h and FV-104 wide open the model reaches 85.5 °C; below the FSL setting, steam on nearly stagnant water boils in the tubes	- FIC-101 LO alarm (80 m³/h) - FSL-101 trip closes XV-100 (< 30 m³/h, 5 s) - TIAH-108 HI alarm (85 °C) - TSHH-108 trip closes XV-100 (> 95 °C, 5 s)	S3 L3 9 Medium
N2.5	Temperature MORE : BFW to the boiler house hotter than intended	- TT-101 reads low - TIC-101 or FIC-104 in MAN at high output - FV-104 fails open - low BFW flow	Hot BFW to the deaerator: flashing, deaerator pressure rise, boiler feed pump cavitation. The model reaches 74.3 °C at the design header and 85.5 °C at 60 m ³ /h, below the TSHH setting	- TIAH-108 HI alarm (85 °C) - TSHH-108 trip closes XV-100 (> 95 °C, 5 s) - TIC-101 HI alarm (74 °C)	S3 L2 6 Medium

HAZOP recommendation

Agree the interface with the boiler house: shared alarms for FIC-101 LO and TIAH-108 HI, a call-out procedure, and include loss of BFW and hot BFW from E-101 in the boiler-house HAZOP.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R08

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R08
Node	N3/N4 · Steam supply: header → XV-100 → FT-104 → FV-104 / E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow MORE : More steam to the shell	- FV-104 fails open - TIC-101 or FIC-104 forced to MAN at high output - FT-104 reads low	Shell pressure rises to about 167 kPa g and the outlet to 74.3 °C at the design header; hot BFW; if the header is also high, see the next row	- TIC-101 HI alarm (74 °C) - FIC-104 HI alarm (13,000 kg/h) - PI-100 HI alarm (200 kPa g) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100	S3 L3 9 Medium
N3.3	Pressure MORE : Shell above its design pressure from the steam side	FV-104 fails open, or the cascade is forced high, while the header is at its highest	Shell pressure rises toward the header pressure; shell or nozzle rupture, large steam release, fatalities nearby. Model: PSHH-100 high shell pressure after 55 s, shell peak 256 kPa g	- PI-100 HI alarm (200 kPa g) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100 (set pressure not on the P&ID) - XV-100 closes on trip	S5 L2 10 High
N4.5	Temperature MORE : External fire on the shell	Fire nearby (oil spill, adjacent unit)	Isolated shell heated: pressure rises, shell rupture	- PSV-100 (fire case to be confirmed) - fire water	S4 L2 8 Medium
N4.8	Integrity OTHER THAN : PSV-100 lifts or passes	- Set too low - seat damage	Steam blowing to atmosphere, noise, scalds at the discharge	Discharge to a safe location	S3 L2 6 Medium

HAZOP recommendation

Confirm the E-101 shell design pressure is at least the highest credible header pressure (144 °C saturation at 300 kPa g in the fault library) and that PSV-100 is set at or below it and sized for FV-104 fully open at that header pressure and for the external-fire case.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R09

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R09
Node	N3 · Steam supply: header → XV-100 → FT-104 → FV-104	Action party	Operations / Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.2	Pressure MORE : Header pressure high	- Upstream flash-plant upset (up to 300 kPa g) - upstream let-down failure	FV-104 throttles to about 30 %; the shell briefly reaches 138 kPa g; control margin at the low end is lost	- PI-105 HI alarm (240 kPa g) - FIC-104 flow control - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100	S3 L3 9 Medium
N3.4	Pressure LESS : Header pressure low	Upstream plant at reduced rate (toward 115 kPa g)	FV-104 wide open; shell 82 kPa g; outlet 68.1 °C, below target (operability)	- PI-105 LO alarm (150 kPa g) - TIC-101 LO alarm (67 °C)	S2 L4 8 Medium

HAZOP recommendation

Header-pressure response procedure with the upstream flash-steam plant for PI-105 LO/HI; consider a pressure-control station on the header so E-101 never sees more than its design pressure.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R10

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R10
Node	N3 · Steam supply: header → XV-100 → FT-104 → FV-104	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.1	Flow MORE : More steam to the shell	- FV-104 fails open - TIC-101 or FIC-104 forced to MAN at high output - FT-104 reads low	Shell pressure rises to about 167 kPa g and the outlet to 74.3 °C at the design header; hot BFW; if the header is also high, see the next row	- TIC-101 HI alarm (74 °C) - FIC-104 HI alarm (13,000 kg/h) - PI-100 HI alarm (200 kPa g) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100	S3 L3 9 Medium
N3.6	Control OTHER THAN : FV-104 stuck	- Actuator or positioner failure - packing friction	Steam does not follow demand: outlet 65.2 °C with the BFW at 120 m³/h in the model; overheating if the BFW falls instead	- TIC-101 LO alarm (67 °C) - TIC-101 HI alarm (74 °C) - operator compares FIC-104 output with the flow	S2 L3 6 Medium
N3.7	Measurement OTHER THAN : FT-104 reads low	Transmitter or impulse-line fault	FIC-104 opens FV-104: transient overheating (outlet up to 74.3 °C) until TIC-101 corrects; steam accounting wrong	- TIC-101 HI alarm (74 °C) - TIAH-108 HI alarm (85 °C)	S2 L3 6 Medium

HAZOP recommendation

Add FV-104 position feedback ZT-104 (recommended addition) with a deviation alarm between FIC-104 output and valve position (stuck or failed valve).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R11

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R11
Node	N4 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.2	Pressure MORE : Shell overpressure from a tube rupture	• Tube rupture (corrosion, vibration, fatigue, water hammer) • BFW side at 662 kPa g (1030 kPa g at shut-off)	Shell fills with water and rises toward the BFW pressure: shell rupture, hot-water and flash-steam release	• PSV-100 (tube-rupture case not confirmed) • PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) stops the steam but not the BFW	S5 L2 10 High

HAZOP recommendation

Apply the tube-rupture (two-thirds) rule: BFW side up to 1030 kPa g at pump shut-off. If the shell design pressure is below two-thirds of the tube-side design pressure, size PSV-100 for the tube-rupture flow of water and flashing water.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R12

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R12
Node	N4/N5 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100 / Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	Action party	Instrument / Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.2	Pressure MORE : Shell overpressure from a tube rupture	• Tube rupture (corrosion, vibration, fatigue, water hammer) • BFW side at 662 kPa g (1030 kPa g at shut-off)	Shell fills with water and rises toward the BFW pressure: shell rupture, hot-water and flash-steam release	• PSV-100 (tube-rupture case not confirmed) • PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) stops the steam but not the BFW	S5 L2 10 High
N4.3	Level MORE : Tube leak into the shell	Pitting, rolled-joint leak	More condensate than steam; hotwell rises to 14.3 % (tubes start at 12 %); steam 9.1 % higher; BFW short at the boiler house; no alarm	• None automatic • condensate bucket test • steam-to-duty comparison	S3 L3 9 Medium
N5.1	Level MORE : Condensate floods the shell (trap or strainer blocked)	• Y-101 screen plugged (scale, debris after maintenance) • trap stuck closed	Lower tubes flooded: outlet falls (40.4 °C after 30 min at 90 % blocking); water hammer when the level reaches about 70 % at 34:43: nozzle or shell damage, steam release, scalds	• TIC-101 LO alarm (67 °C) only • E-101 HAMMER event alarm (after the fact) • no level instrument	S4 L3 12 High
N5.3	Pressure LESS : Shell pressure below the return header	• Low steam demand (warm BFW, low BFW flow) or low header pressure throttles the shell • shell under vacuum after a trip	Trap has no driving pressure: condensate backs up and floods the lower tubes (the N2 low-flow run on the simulator floods the hotwell), hammer risk when the load returns	• TIC-101 LO alarm (67 °C) (late) • no level instrument	S3 L3 9 Medium

HAZOP recommendation

Add a shell hotwell level transmitter LT-100 (recommended addition) with a high alarm below the tube bottom (12 % of the shell) and a high-high level trip closing XV-100 before steam can meet a flooded shell.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R13

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R13
Node	N4 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	Action party	Process
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.6	Temperature LESS : BFW to the boiler house colder than intended	• Low header pressure • gas blanketing • condensate flooding • tube scale • FV-104 stuck	Boiler house needs more heat; deaerator steam demand up (operability)	TIC-101 LO alarm (67 °C)	S2 L4 8 Medium
N4.1	Composition AS WELL AS : Non-condensable gas in the shell	• Gas carried in the flash steam • air drawn in through VB-101 not vented at restart • vent closed	Blanketed surface: shell pressure creeps up (172 kPa g after 30 min) with no alarm; outlet 65.3 °C after 1 h and 56.7 °C after 2 h; CO ₂ gives corrosive condensate	• V-104 manual vent • TIC-101 LO alarm (67 °C) (late) • rev B / rev C not in service	S2 L4 8 Medium
N6.3	Venting PART OF : Vent closed before dry steam blows	V-104 closed too early at start-up	Air left in the shell: surface blanketed, FV-104 wide open, cannot reach 70 °C	• Procedure step "vent until dry steam blows" • TIC-101 LO alarm (67 °C)	S2 L4 8 Medium

HAZOP recommendation

Put a continuous vent in service (rev B RO-101) or the automatic air vent with vent-line temperature (rev C AAV-101, TI-102, TAL-102): gas blanketing gives no alarm for a long time.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R14

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R14
Node	N4 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	Action party	Mechanical
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.5	Flow NO : No steam	• XV-100 closed (trip, spurious trip, loss of air or power) • FV-104 fails closed • header lost	Cold BFW to the boiler house; shell drawn to vacuum (–3 kPa g in the model) and air enters through VB-101; restart without venting leaves the shell blanketed	• VB-101 • TIC-101 LO alarm (67 °C) • restart procedure	S3 L3 9 Medium
N4.4	Pressure LESS : Shell under vacuum	Steam cut (trip, ESD, FV-104 closed) while cold BFW flows	Shell falls to –3 kPa g within seconds; shell buckling if VB-101 fails; trap cannot discharge; air enters	VB-101	S3 L2 6 Medium

HAZOP recommendation

Confirm the shell is rated for full vacuum or that VB-101 is sized for the fastest condensation after a trip; add VB-101 to the preventive-maintenance and test plan.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R15

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R15
Node	N5 · Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	Action party	Mechanical / Operations
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.6	Temperature LESS : BFW to the boiler house colder than intended	- Low header pressure - gas blanketing - condensate flooding - tube scale - FV-104 stuck	Boiler house needs more heat; deaerator steam demand up (operability)	TIC-101 LO alarm (67 °C)	S2 L4 8 Medium
N5.1	Level MORE : Condensate floods the shell (trap or strainer blocked)	- Y-101 screen plugged (scale, debris after maintenance) - trap stuck closed	Lower tubes flooded: outlet falls (40.4 °C after 30 min at 90 % blocking); water hammer when the level reaches about 70 % at 34:43: nozzle or shell damage, steam release, scalds	- TIC-101 LO alarm (67 °C) only - E-101 HAMMER event alarm (after the fact) - no level instrument	S4 L3 12 High
N5.3	Pressure LESS : Shell pressure below the return header	- Low steam demand (warm BFW, low BFW flow) or low header pressure throttles the shell - shell under vacuum after a trip	Trap has no driving pressure: condensate backs up and floods the lower tubes (the N2 low-flow run on the simulator floods the hotwell), hammer risk when the load returns	- TIC-101 LO alarm (67 °C) (late) - no level instrument	S3 L3 9 Medium
N5.4	Flow MORE : Trap passes live steam (fails open)	Worn trap valve or seat	Steam 18.1 % higher for the same duty; return-header pressure up, hammer in the return line; scalds at leaks	- None automatic - trap surveys	S3 L3 9 Medium
N5.5	Flow AS WELL AS : Bypass V-103 left open	Left open after a trap problem	Live steam lost (7.7 % more steam in the model); flashing in the return header	- Procedure - trap-set rounds	S2 L3 6 Medium
N5.7	Flow REVERSE : Back-flow from the return header	Shell under vacuum after a trip while the header is at 30 kPa g	Condensate from the header back into the shell: flooding, hammer at restart	Trap resists reverse flow (check valve not shown)	S3 L2 6 Medium

HAZOP recommendation

Trap management: regular trap survey (ultrasonic or temperature), Y-101 cleaning after any maintenance, and confirm ST-101 capacity at low shell differential pressure (start-up and near-vacuum operation).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R16

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R16
Node	N5 · Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	Action party	Process / Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N5.2	Pressure MORE : Return header back-pressure high	- Other users or flash in the common header - header valve partly closed	Trap discharges only when the shell is above the header: shell floods (level 39 %), outlet 59.9 °C, shell 182 kPa g; hammer risk	- Local gauge PG-110 - TIC-101 LO alarm (67 °C)	S3 L3 9 Medium
N5.6	Flow OTHER THAN : Hot condensate drained to sewer	V-105 opened with the shell under pressure	Condensate at 119 °C flashes in the sewer: steam cloud, scalds, damage to the sewer, hot discharge (3.3 % more steam while open)	Procedure: drain only after the shell pressure has fallen	S3 L3 9 Medium

HAZOP recommendation

Replace local gauge PG-110 by a return-header pressure transmitter with a high alarm (recommended addition); route shell drain V-105 to a flash pot or blowdown pit, not to an open sewer.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R17

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R17
Node	N4/N5 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100 / Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	Action party	Mechanical / Safety
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.7	Flow MORE : Vent V-104 left wide open	Not closed after start-up	Live steam lost (9.1 % more steam in the model); noise and scalding plume at the discharge	• Procedure (close to 10 % or less) • rev B / rev C	S2 L3 6 Medium
N4.8	Integrity OTHER THAN : PSV-100 lifts or passes	• Set too low • seat damage	Steam blowing to atmosphere, noise, scalds at the discharge	Discharge to a safe location	S3 L2 6 Medium
N5.6	Flow OTHER THAN : Hot condensate drained to sewer	V-105 opened with the shell under pressure	Condensate at 119 °C flashes in the sewer: steam cloud, scalds, damage to the sewer, hot discharge (3.3 % more steam while open)	Procedure: drain only after the shell pressure has fallen	S3 L3 9 Medium

HAZOP recommendation

Route V-104, PSV-100, V-103 and V-105 discharges to a safe location; fit a silencer on the vent; warning signs and hearing-protection zone around the vent and trap set.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R18

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R18
Node	N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.2	Warm-up EARLY : Steam raised early, before the shell is warm	- FIC-104 opened too far at start-up - the permissive checks FIC-104 only at the moment XV-100 opens	Water hammer: at 10 % output the model bangs 6.8 s after XV-100 opens in a cold shell (1.8 s at 40 %; none at 5 %); gasket or nozzle failure, scalds	- Start permissive FIC-104 in MAN $\leq$ 10 % - warm-up step in the procedure - E-101 HAMMER alarm	S4 L3 12 High
N6.3	Venting PART OF : Vent closed before dry steam blows	V-104 closed too early at start-up	Air left in the shell: surface blanketed, FV-104 wide open, cannot reach 70 °C	- Procedure step "vent until dry steam blows" - TIC-101 LO alarm (67 °C)	S2 L4 8 Medium
N6.7	ESD LATE : Steam not stopped quickly on an emergency order	Operator unfamiliar with the SIS screen	Hot BFW or steam keeps flowing during a boiler-house emergency	- ESD pushbutton on the SIS screen - procedure	S3 L2 6 Medium

HAZOP recommendation

Start-up procedure hold points: BFW in AUTO first, vent V-104 open, warm-up with FIC-104 below 10 % (the model gives water hammer at 10 % in a cold shell, none at 5 %) until the shell is warm, vent until dry steam blows, then cascade.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R19

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R19
Node	N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N2.4	Pressure MORE : Tube side blocked in while hot	Inlet and outlet isolated for maintenance with steam still on or the shell still at 121.3 °C	Liquid thermal expansion overpressures the tube side: gasket or flange leak of hot water, tube damage	None on the P&ID (no tube-side relief)	S4 L2 8 Medium
N6.5	Sequence AFTER : Steam isolated after the BFW is stopped	Pump stopped first at shutdown	Steam on stagnant water: boiling in the tubes	- FSL-101 trip closes XV-100 (< 30 m³/h, 5 s) - shutdown procedure	S3 L2 6 Medium
N6.6	Sequence BEFORE : Drain opened before the shell is depressured	Hurry at shutdown	Flashing condensate at the drain: scalds	Procedure: open V-105 once the shell pressure has fallen	S3 L2 6 Medium

HAZOP recommendation

Shutdown procedure: steam off and shell depressured before the BFW is stopped or blocked in; open V-105 only after the shell pressure has fallen; keep V-104 open during lay-up.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R20

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R20
Node	N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	Action party	Operations / Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.1	Sequence BEFORE : Steam admitted before the BFW flows	- Operator error - permissive bypassed	Steam on stagnant water: boiling in the tubes, overpressure, hammer	- XV-100 start permissive BFW ≥ 50 m³/h - FSL-101 trip closes XV-100 (< 30 m³/h, 5 s)	S4 L1 4 Low
N6.9	Safety system OTHER THAN : Trip bypassed or overridden	Testing, nuisance trips	No protection against low flow, high outlet temperature or high shell pressure	Permit to work	S4 L2 8 Medium

HAZOP recommendation

Trip-bypass management for FSL-101, TSHH-108 and PSHH-100: permit, time limit, MOC, bypass status shown on the DCS; proof-test the XV-100 trips at the interval set by LOPA.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R21

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R21
Node	N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	Action party	Operations / Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.5	Flow NO : No steam	• XV-100 closed (trip, spurious trip, loss of air or power) • FV-104 fails closed • header lost	Cold BFW to the boiler house; shell drawn to vacuum (–3 kPa g in the model) and air enters through VB-101; restart without venting leaves the shell blanketed	• VB-101 • TIC-101 LO alarm (67 °C) • restart procedure	S3 L3 9 Medium
N6.8	Venting NO : Restart without venting	SIS reset and steam readmitted with V-104 closed after a trip (air drawn in through VB-101)	Blanketed shell, low duty, cold BFW to the boiler house	Restart procedure	S2 L4 8 Medium

HAZOP recommendation

Restart after any trip: re-vent the shell (air drawn in through VB-101) before the cascade is put back; consider "V-104 open" as a start permissive.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R22

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R22
Node	N6 · Procedures: start-up venting and warm-up, shutdown, ESD, restart; utilities	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N6.10	Utility NO : Loss of instrument air	Compressor trip, air-line break	FV-104 fails closed and XV-100 closes (steam off, shell to vacuum); FV-101A fails open (pump runs out); FV-101B position not stated	• Fail positions: FV-104 FC, FV-101A FO • VB-101	S3 L3 9 Medium
N6.11	Utility NO : Loss of power	Grid or substation failure	Pumps stop: FSLL-101 closes XV-100; no BFW to the boiler house	• FSLL-101 trip closes XV-100 (< 30 m³/h, 5 s) • P-101B (on the same supply)	S4 L2 8 Medium

HAZOP recommendation

Confirm the fail position of FV-101B (not shown on the P&ID) and that XV-100 closes on loss of air or power; document the fail positions in the cause-and-effect chart.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R23

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R23
Node	all · all	Action party	Process safety
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.3	Pressure MORE : Shell above its design pressure from the steam side	FV-104 fails open, or the cascade is forced high, while the header is at its highest	Shell pressure rises toward the header pressure; shell or nozzle rupture, large steam release, fatalities nearby. Model: PSHH-100 high shell pressure after 55 s, shell peak 256 kPa g	- PI-100 HI alarm (200 kPa g) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) - PSV-100 (set pressure not on the P&ID) - XV-100 closes on trip	S5 L2 10 High
N4.2	Pressure MORE : Shell overpressure from a tube rupture	- Tube rupture (corrosion, vibration, fatigue, water hammer) - BFW side at 662 kPa g (1030 kPa g at shut-off)	Shell fills with water and rises toward the BFW pressure: shell rupture, hot-water and flash-steam release	- PSV-100 (tube-rupture case not confirmed) - PSHH-100 trip closes XV-100 (> 250 kPa g, 3 s) stops the steam but not the BFW	S5 L2 10 High

HAZOP recommendation

LOPA for the High rows that rely on PSHH-100, FSLL-101 and PSV-100 (shell overpressure, tube rupture, loss of BFW, water hammer).

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R24

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R24
Node	N3 · Steam supply: header → XV-100 → FT-104 → FV-104	Action party	Instrument
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.8	Temperature MORE : Superheated steam	Let-down upstream without desuperheating (up to 40 °C superheat)	Hotter inlet nozzle and surfaces; a dry-wall zone at the inlet; shell runs higher (143 kPa g in the model)	TI-100 read against the saturation temperature of PI-100	S2 L2 4 Low

HAZOP recommendation

Show a calculated superheat TI-100 – Tsat(PI-100) on the DCS; tell the upstream plant when it exceeds a few degrees.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R25

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R25
Node	N1/N2 · BFW supply: TK-101 → P-101A/B → recycle FV-101B / BFW flow control FIC-101 / FV-101A, E-101 tube side, delivery to the boiler house	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N1.10	Composition AS WELL AS : Oxygen, hardness or wrong treatment chemical in the BFW	- Dosing upset - poor make-up quality	Corrosion or scale in the tubes (tube leak later), boiler deposits; exposure when sampling	Water-quality sampling programme	S2 L3 6 Medium
N2.8	Heat transfer LESS : Scale inside the tubes	- Hardness in the BFW - poor treatment	Duty lost, FV-104 wide open, outlet falls (54.8 °C at full scale in the model); higher tube ΔP	- PDI-109 HI alarm (80 kPa) - TIC-101 LO alarm (67 °C)	S2 L3 6 Medium

HAZOP recommendation

Water-treatment control: BFW hardness and oxygen-scavenger sampling programme; closed sampling point and PPE for treatment chemicals.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R26

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R26
Node	N3 · Steam supply: header → XV-100 → FT-104 → FV-104	Action party	Mechanical / Operations
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N3.9	Composition AS WELL AS : Condensate slugs in the steam branch	- No drip leg upstream of XV-100 - cold line at start-up	Water hammer in the steam line: damage to FV-104, FT-104, flanges; steam release	None shown (line drainage upstream not on the P&ID)	S4 L2 8 Medium

HAZOP recommendation

Confirm a drip leg and trap on the steam branch upstream of XV-100, and warm the line with its drains open before admitting steam.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R27

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R27
Node	N4 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100	Action party	Operations
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.6	Heat transfer LESS : Oil film on the tubes (shell side)	Oil carried in the flash steam from upstream	Duty falls (54.8 °C at full severity), PDI-109 unchanged; oil in the condensate to the boiler house	- TIC-101 LO alarm (67 °C) - condensate appearance	S2 L3 6 Medium

HAZOP recommendation

Condensate quality: oil-in-condensate check or monitor on the return to the boiler house; find and stop the oil source in the flash steam.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			

HAZOP ACTION CLOSE-OUT SHEET · R28

Project	BFW Steam Heater Ver 1 · E-101 Operator Training Simulator	Document	BFWH-V1-SA-HZP-001 rev A
Engineering phase	Teaching study (basic design level)	Action No.	R28
Node	N4/N5 · E-101 shell: condensing space with vent V-104, VB-101 and PSV-100 / Condensate: hotwell → V-101 / Y-101 / ST-101 / V-102 → return header; V-103 bypass, V-105 drain	Action party	Operations / Maintenance
Drawings	P&ID-001 rev C (screen drawing); BFWH-V1-PID-001 rev D (alternative EPC drawing)	Due date	

Ref	Deviation	Causes	Consequences	Existing safeguards	Risk
N4.9	Maintenance OTHER THAN : Shell entry or bundle pull	Inspection, tube plugging	Scalds from trapped steam or hot water, oxygen-deficient or hot atmosphere in the shell, dropped bundle	• Permit to work • isolation of steam, BFW and condensate • lifting procedure	S4 L2 8 Medium
N5.9	Maintenance OTHER THAN : Opening the strainer or trap while hot	• Block valves V-101 / V-102 passing • line not drained	Scalds from hot condensate and flash steam	• Permit to work • isolation and drain	S3 L3 9 Medium

HAZOP recommendation

Maintenance isolation procedure for E-101: positive isolation (blinds) of steam, BFW and condensate, drain and cool before opening, confined-space permit for shell entry, lift plan for bundle pulls.

Response of the action party (what was done, or the alternative and its justification, or the reason for rejection)

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Status	☐ Accepted and done	☐ Accepted with alternative	☐ Rejected	☒ Open (at issue)
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Sign-off	Name	Signature	Date
Action party			
HAZOP leader			
Plant / project manager			